PRACTICE SHEET

1. $A = \begin{vmatrix} 2a & 3r & x \\ 4b & 6s & 2y \\ -2c & -3t & -z \end{vmatrix} = \lambda \begin{vmatrix} a & r & x \\ b & s & y \\ c & t & z \end{vmatrix}$, then what is the value of λ?
(a) 12 (b) –12
(c) 7 (d) –7

2. What is the value of $\begin{vmatrix} 1-i & \omega^2 & -\omega \\ \omega^2+i & \omega & -i \\ 1-2i-\omega^2 & \omega^2-\omega & i-\omega \end{vmatrix}$, where ω is the rube root of unity?
(a) –1 (b) 1
(c) 2 (d) 0

3. If a, b and c are non-zero real numbers and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, then what is the value of $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$?
(a) 2 (b) 1
(c) –1 (d) 0

4. If $\begin{vmatrix} y & x & y+z \\ z & y & x+y \\ x & z & z+x \end{vmatrix}$ =0, then which one of the following is correct?
(a) Either x + y = z or x = y
(b) Either x + y = – z or x = z
(c) Either x + z = y or z = y
(d) Either z + y = x or x = y

5. What is the value of k, if $\begin{vmatrix} k & b+c & b^2+c^2 \\ k & c+a & c^2+a^2 \\ k & a+b & a^2+b^2 \end{vmatrix}$ =(a–b) (b–c) (c–a)?
(a) 1 (b) –1
(c) 2 (d) 0

6. What is the value of $\begin{vmatrix} \sin 10^\circ - & \cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix}$?
(a) 0 (b) 1
(c) –1 (d) $\frac{1}{2}$

7. If $\begin{vmatrix} 2 & 4 & 0 \\ 0 & 5 & 16 \\ 0 & 0 & 1+p \end{vmatrix}$ =20, then what is the value of p?
(a) 0 (b) 1
(c) 2 (d) 5

8. If $(a_1 / x) + b_1 / y)\, c_1, (a_2 / x) + b_2 / y) = c_2$

$\Delta_1 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \Delta_2 = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}, \Delta_3 \begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}$,
Then (x, y) is equal to which one of the following?
(a) $(\Delta_2/\Delta_1, \Delta_3/\Delta_1)$ (b) $(\Delta_3/\Delta_1, \Delta_2/\Delta_1)$
(c) $(\Delta_1/\Delta_2, \Delta_1/\Delta_3)$ (d) $(-\Delta_1/\Delta_2, \Delta_1/\Delta_3)$

9. What is the determinant $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$ equal to?
(a) $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ (b) $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b_2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$
(c) $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$ (d) $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

10. If $x^2 + y^2 + z^2 = 1$, then what is the value of $\begin{vmatrix} 1 & z & -y \\ -z & 1 & x \\ y & -x & 1 \end{vmatrix}$?
(a) 0 (b) 1
(c) 2 (d) 2–2xyz

11. If $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$
What is the maximum value of f x?
(a) 2 (b) 4
(c) 6 (d) 8

12. For what values of k, does the system of linear equations x + y + z = 2, 2x + y – z = 3, 3x +2y + kz = 4 have a unique solution?
(a) k = 0 (b) $-1 < k < 1$
(c) $-2 < k < 2$ (d) $k \neq 0$

13. If a matrix B is obtained from a square matrix A by interchanging any two of its rows, then what is |A+B| equal to
(a) 2|A| (b) 2|B|
(c) 0 (d) |A|–|B|

14. What is the largest value of a third order determinant whose elements are 0 or 1?
(a) 0 (b) 1
(c) 2 (d) 3

15. What is the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ if $a^3+b^3+c^3=0$?
(a) 0 (b) 1
(c) 3 abc (d) –3 abc

16. If $\begin{vmatrix} x & x^2 & 1+x^2 \\ y & y^2 & 1+y^2 \\ z & z^2 & 1+z^2 \end{vmatrix}$ where x, y, z are distinct what is |A|?

(a) 0
(b) $x^2y - y^2x + xyz$
(c) $(x - y)(y - z)(z - x)$
(d) xyz

17. Consider the following statements:
1. If det A = 0, then det (adj A) = 0
2. If A is non – singular, then det $(A^{-1}) = (\det A)^{-1}$

(a) 1only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

18. Let A be a square matrix of order $n \times n$ where $n \geq 2$. Let B be a matrix obtained from A with first and second rows interchanged. Then which one of the following is correct?
(a) det A = det B (b) det A = – det B
(c) A = B (d) A = – B

19. What should be the value of k so that the system of linear equations $x - y + 2z = 0$, $kx - y + z = 0$, $3x + y - 3z = 0$ does not possess a unique solution?
(a) 0 (b) 3
(c) 4 (d) 5

20. For how many values of k, will the system of equations $(k + 1)x + 8y = 4k$ and $kx + (k + 3)y = 3k - 1$, have an infinite number of solutions
(a) 1 (b) 2
(c) 3 (d) None of these

21. If $\begin{vmatrix} a & b & c \\ l & m & n \\ p & q & r \end{vmatrix} = 2$, then what is the value of the determinant $\begin{vmatrix} 6a & 3b & 15c \\ 2l & m & 5n \\ 2p & q & 5r \end{vmatrix} = ?$
(a) 10 (b) 20
(c) 40 (d) 60

ANSWER KEY																			
1.	b	**2.**	d	**3.**	c	**4.**	b	**5.**	a	**6.**	b	**7.**	b	**8.**	d	**9.**	b	**10.**	c
11.	c	**12.**	d	**13.**	c	**14.**	c	**15.**	c	**16.**	c	**17.**	c	**18.**	b	**19.**	d	**20.**	c
21.	d																		

Solutions

Sol.1. (b)
Take 2 common from first column
Take 3 common from second column
Take 2 common from second row
Take -1 common from third row
$\lambda = -12$

Sol.2. (d)

$$\begin{vmatrix} 1-i & \omega^2 & -\omega \\ \omega^2+i & \omega & -i \\ 1-2i-\omega^2 & \omega^2-\omega & i-\omega \end{vmatrix}$$

Apply $R_3 \leftrightarrow R_1 - R_2 - R_3$

$$\begin{vmatrix} 1-i & \omega^2 & -\omega \\ \omega^2+i & \omega & -i \\ 0 & 0 & 0 \end{vmatrix} = 0$$

Sol.3. (c)

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$$

Applying $C_2 \leftrightarrow C_2 - C_2$ and $C_3 \leftrightarrow C_3 - C_1$

$$\begin{vmatrix} 1+a & -a & -a \\ 1 & b & 0 \\ 1 & 0 & c \end{vmatrix} = 0$$

Now after expanding $ab + bc + ca + abc = 0$

Sol.4. (b)

$$\begin{vmatrix} y & x & y+z \\ z & y & x+y \\ x & z & z+x \end{vmatrix} = 0$$

Apply $R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} x+y+z & x+y+z & 2(x+y+z) \\ z & y & x+y \\ x & z & z+x \end{vmatrix} = 0$$

$$(x+y+z)\begin{vmatrix} 1 & 1 & 2 \\ z & y & x+y \\ x & z & z+x \end{vmatrix} = 0$$

Applying $C_2 \leftrightarrow C_1 - C_2$ and $C_3 \leftrightarrow C_3 - 2C_1$

$$(x+y+z)\begin{vmatrix} 1 & 0 & 0 \\ z & z-y & x+y-2z \\ x & z-x & z-x \end{vmatrix} = 0$$

$(x + y + z)(z - x) = 0$

$x + y = -z$ or $z = x$

Sol.5. (a)

Let $\Delta \begin{vmatrix} k & b+c & b^2+c^2 \\ k & c+a & c^2+a^2 \\ k & a+b & a^2+b^2 \end{vmatrix}$

$$=k\begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ k & a+b & a^2+b^2 \end{vmatrix}$$

$$=k\begin{vmatrix} 1 & 0 & 0 \\ b+c & a-b & a-c \\ b^2+c^2 & a^2-b^2 & a^2-c^2 \end{vmatrix}$$

$\left(\begin{matrix} \because \quad \ldots\ _2 - C_1 \\ \text{and } C_3 \to C_3 - C_1 \end{matrix}\right)$

$=k(a-b)(a-c)\begin{vmatrix} 1 & 1 \\ a+b & a+c \end{vmatrix}$

$=k(a - b)(a - c)(a + c - a - c)$
$= k(a - b)(b - c)(c - a)$
But $\Delta = (a - b)(b - c)(c - a)$
Thus, $k = 1$

Sol.6. (b)

$$\begin{vmatrix} \sin 10^\circ & -\cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix} = \begin{vmatrix} \sin 10^\circ & -\cos 10^\circ \\ \cos 10^\circ & \sin 10^\circ \end{vmatrix}$$

$\sin^2 10^\circ + \cos^2 10^\circ = 1$

Sol.7. (b)
$10(1 + p) = 20$
$1 + p = 2$
$p = 1$

Sol.8. (d)

Let $\frac{1}{x} = u, \frac{1}{y} = v$

$\therefore a_1u + b_1v = c_1$ and $a_2u + b_2v = c_2$
Using the method of cross multiplication,

$$\frac{u}{b_1c_2 - b_2c_1} = \frac{v}{c_1a_2 - c_2a_1} = \frac{-1}{a_1b_2 - a_2b_1}$$

$$\Rightarrow \frac{\frac{1}{x}}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{\frac{1}{y}}{\begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}} = \frac{-1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

$$\Rightarrow \frac{\frac{1}{x}}{\Delta_2} = \frac{\frac{1}{y}}{\Delta_3} = \frac{-1}{\Delta_1}$$

$$\therefore \frac{1}{x} = \frac{\Delta_2}{\Delta_1} \text{ and } \frac{1}{y} = -\frac{\Delta_3}{\Delta_1}$$

$$\Rightarrow x = -\frac{\Delta_1}{\Delta_2} \text{ and } y = -\frac{\Delta_1}{\Delta_3}$$

Sol.9. (b)

$$\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$$

$R_1 \to aR_1$, $R_2 \to aR_2$, $R_3 \to aR_3$ and divide whole determinant by abc.

$$\frac{1}{abc}\begin{vmatrix} abc & a^2 & a^3 \\ abc & b^2 & b^3 \\ abc & c^2 & c^3 \end{vmatrix}$$

$$\frac{abc}{abc}\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

Sol.10. (c)

After expanding $(1 + x^2) - z(-z - xy) - y(xz - y)$

$1 + x^2 + y^2 + z^2 = 2$

Sol.11. (c)

$$f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

Applying $C_1 \to C_1 + C_2$

$$= \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 2 & 1+\cos^2 x & 4\sin 2x \\ 1 & \cos^2 x & 1+4\sin 2x \end{vmatrix}$$

(Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$)

$$= \begin{vmatrix} 2 & \cos^2 x & 4\sin 2x \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix}$$

$f(x) = 2 + 4\sin 2x$

$\therefore -1 \le \sin 2x \le 1$,

maximum value of $\sin 2x = 1$

Thus, maximum value of $f(x) = 2 + 4 = 6$

Sol.12. (d)

The given system of equation is

$x + y + z = 2$

$2x + y - z = 3$

And $3x + 2y + kz = 4$

This system has a unique solution if

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & k \end{vmatrix} \ne 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

We get $\begin{vmatrix} 1 & 0 & 0 \\ 3 & -1 & -3 \\ 3 & -1 & k-3 \end{vmatrix} \ne 0$

$\Rightarrow 1(k-3) - 3 \ne 0$ or $-k + 3 - 3 \ne 0$

$\Rightarrow k \ne 0$

Sol.13. (c)

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

Let rows 1 and 2 be interchanged.

and $B = \begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix}$

$$A+B = \begin{bmatrix} a+d & b+e & c+f \\ a+d & b+e & c+f \\ 2g & 2h & 2i \end{bmatrix}$$

=0 (since two rows are identical)

Sol.14. (c)

In the third order determinant $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$ which satisfied these conditions. Its value is 2, which is largest if the elements of a determinant are 0 r 1.

Sol.15. (c)

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

$= a(bc - a^2) - b(b^2 - ac) + c(ab - c^2)$

$= abc - a^3 - b^3 + abc + abc - c^3 = abc - (a^3 + b^3 + c^3)$

Given that $a^3 + b^3 + c^3 = 0$

$$\Rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc$$

Sol.16. (c)

$$A = \begin{bmatrix} x & x^2 & 1+x^2 \\ y & y^2 & 1+y^2 \\ z & z^2 & 1+z^2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} x & x^2 & 1+x^2 \\ y & y^2 & 1+y^2 \\ z & z^2 & 1+z^2 \end{vmatrix}$$

Applying $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$

$$|A| = \begin{vmatrix} x-y & (x-y)(x+y) & (x-y)(x+y) \\ y-z & (y-z)(y+z) & (y-z)(y+z) \\ z & z^2 & 1+z^2 \end{vmatrix}$$

$$=(x-y)(y-z)\begin{vmatrix} 1 & x+y & x+y \\ 1 & y+z & y+z \\ z & z^2 & 1+z^2 \end{vmatrix}$$

Applying $C_3 \to C_3 - C_2$

$$|A| = (x-y)(y-z)\begin{vmatrix} 1 & x+y & 0 \\ 1 & y+z & 0 \\ z & z^2 & 1 \end{vmatrix}$$

$=(x-y)(y-z)[1+\{y+z-(x+y)\}]$

$=x(x-y)(y-z)(z-x)$

Sol.17. (c)

We know that, adj A and A has same value of determinant, if det A = 0, then det (adJ A) = 0

So, statement (1) is correct.

Also if A is a matrix the determinant of A^{-1} equals inverse of determinant A, so, det (A^{-1}) = $(\det A)^{-1}$, if A is non singular ; Statement 2 is correct.

Thus both (1) and (2) are correct.

Sol.18. (b)

A be a square matrix of order $n \times n$ where $n \ge 2$. B be a matrix obtained from A with first and second rows interchanged. Then, det A = – det B. Since interchanging any two rows makes the sign change with same value.

Sol.19. (d)

Equations are

$x - y + 2x = 0$ ...... (i)

$kx - y + z = 0$ ....(ii)

$3x + y - 3x = 0$ ....(iii)

System of equations posses a unique solution, if

$$|A| = \begin{vmatrix} 1 & -1 & 2 \\ k & -1 & 1 \\ 3 & 1 & -3 \end{vmatrix} \ne 0$$

Applying $C_1 \to C_1 + C_2$ and $C_2 \to C_2 + 1/2\, C_3$

$$|A| = \begin{vmatrix} 1 & 0 & 0 \\ k-1 & -1/2 & 1 \\ 4 & -1/2 & -3 \end{vmatrix} = -1/2\begin{vmatrix} 0 & 0 & 2 \\ k-1 & -1 & 1 \\ 4 & -1 & -3 \end{vmatrix}$$

$=1/2 \times 2\,[(l-1)(-1)-(4)(-1)],0$

$\Rightarrow -(-k+1+4) \ne 0 \Rightarrow k-5 \ne 0 \Rightarrow k \ne 5$

Thus, he system does not possess unique solution, if k = 5

Sol.20. (c)

System of equation is given as:

$(k + 1)x + 8y = 4k$ ......(1)

And $kx + (k + 3)y = 3k - 1$ .....(2)

Here, $a_1 = k + 1$, $b_1 = 8$, $c_1 = 4k$, $a_2 = k$, $b_2 = k + 3$ and $c_2 = 3k - 1$

Such a system of equations will have infinite number of solution, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

i.e., $\frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$

Taking last two we get

$8(3k - 1) = 4k(k + 3)$

$\Rightarrow 24k - 8 = 4k^2 + 12k$

$\Rightarrow 4k^2 - 12k + 8 = 0$

$\Rightarrow k^2 - 3k + 2 = 0$

$\Rightarrow (k-1)(k-2) = 0$

$\Rightarrow k = 1, 2$

Taking first two $(k + 1)(k + 3) = 8k$

$\Rightarrow k^2 + 4k + 3 - 8k = 0 \Rightarrow k^2 - 4k + 3 = 0$

$\Rightarrow (k - 1)(k - 3) = 0$

So, k = 1, 3

Combining both, k = 1, 2, 3

Thus, this system has 3 values of k.

Sol.21. (d)

$$\begin{vmatrix} 6a & 3b & 15c \\ 2l & m & 5n \\ 2p & q & 5r \end{vmatrix} = 30\begin{vmatrix} a & b & c \\ l & m & n \\ p & q & r \end{vmatrix} = 30 \times 2 = 60$$

NDA PYQ

1. What is the value of the determinant?
$$\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}?$$
(a) x+2 (b) $x^2 + 2$
(c) 2 (d) –2

[NDA (I) - 2011]

2. If 5 and 7 are the roots of the equation $\begin{vmatrix} x & 4 & 5 \\ 7 & x & 7 \\ 5 & 8 & x \end{vmatrix} = 0$, then what is the third root?
(a) –12 (b) 9
(c) 13 (d) 14

[NDA (I) - 2011]

3. What is the value of k for which the system of equations kx+2y=5 and 3x+y=1 has no solution.
(a) 0 (b) 3
(c) 6 (d) 15

[NDA-2011(1)]

4. The root of the equation $\begin{vmatrix} x & \alpha & 1 \\ \beta & x & 1 \\ \beta & \gamma & 1 \end{vmatrix}$=0 are independent of:
(a) α (b) β
(c) γ (d) α, β and γ

[NDA (II) - 2011]

5. If $\begin{vmatrix} p & -q & 0 \\ 0 & p & q \\ q & 0 & p \end{vmatrix}$=0, then which one of the following is correct?
(a) p is one of the cube roots of unity
(b) q is one of the cube roots of unity
(c) $\frac{p}{q}$ is one of the cube roots of unity
(d) None of the above

[NDA (II) - 2011]

6. What is the value of the determinant $\begin{vmatrix} a-b & b+c & a \\ b-c & c+a & b \\ c-a & a+b & c \end{vmatrix}$?
(a) $a^3 + b^3 + c^3$ (b) 3bc
(c) $a^3 + b^3 + c^3 - 3abc$ (d) 0

[NDA (II) - 2011]

7. The simultaneous equations 3x+5y=7 and 6x+10y=18 have:
(a) no solution
(b) infinitely many solution
(c) unique solution
(d) any finite number of solutions

[NDA-2011(II)

8. If $a^{-1} + b^{-1} + c^{-1} = 0$ such that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = \lambda$, then what is λ equal to?
(a) –abc (b) abc
(c) 0 (d) 1

[NDA-2011(II)]

9. Consider the following statements in respect of the square matrices A and B of same order:
1. A and B are non-zero and AB = 0 $\Rightarrow$ either |A|=0 or |B|=0
2. AB=0$\Rightarrow$A=0 or B=0
Which of the above statements is/are correct?
(a) 1 only (b) 2only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2011(II)]

10. If two rows of a determinant are identical, then what is the value of the determinant?
(a) 0
(b) 1
(c) –1
(d)Can be any real value

[NDA (I) - 2012]

11. If $\begin{vmatrix} 8 & -5 & 1 \\ 5 & x & 1 \\ 6 & 3 & 1 \end{vmatrix} = 2$, then what is the value of x?
(a) 4 (b) 5
(c) 6 (d) 8

[NDA (I) - 2012]

12. If $A = \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}$ and $B = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}$ then what is the value of determinant of AB?
(a) 0 (b)1
(c) 10 (d) 20

[NDA (I) - 2012]

13. What is the value of $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix}$?
(a) 4abc (b) $4a^2bc$
(c) $4a^2b^2c^2$ (d) $-4a^2b^2c^2$

[NDA (I) - 2012]

14. If each element in a row of a determinant is multiplied by the same factor r, then the value of the determinant
(a) Is multiplied by r^3 (b) Is increased by 3r
(c) Remains unchanged (d) Is multiplied by r

[NDA (II) - 2012]

15. The value of the determinant $\begin{vmatrix} x^2 & 1 & y^2+z^2 \\ y^2 & 1 & z^2+x^2 \\ z^2 & 1 & x^2+y^2 \end{vmatrix}$ is:
(a) 0 (b) $x^2 + y^2 + z^2$
(c) $x^2 + y^2 + z^2 + 1$ (d) none of the above

[NDA-2012(2)]

16. If D is determinant of order 3 and D' is the determinant obtained by replacing the elements of D by their cofactors, then which one of the following is correct?
(a) $D' = D^2$ (b) $D' = D^3$
(c) $D' = 2D^2$ (d) $D' = 3D^3$

[NDA (I) - 2013]

17. Consider the following statements
I. A matrix is not a number.

II.Two determinants of different orders may have the same value.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2013]

18. The roots of the equation $\begin{vmatrix} 1 & t-1 & 1 \\ t-1 & 1 & 1 \\ 1 & 1 & t-1 \end{vmatrix} = 0$ are
(a) 1, 2 (b) –1, 2
(c) 1, –2 (d) –1, –2
[NDA (I) - 2013]

19. The value of the determinant $\begin{vmatrix} m & n & p \\ p & m & n \\ n & p & m \end{vmatrix}$
(a) Is a perfect cube (b) Is a perfect square
(c) Has linear factor (d) Is zero
[NDA (I) - 2013]

20. For what value of k, the equations 3x–y=8 and 9x–ky=24 will have infinitely many solutions?
(a) 6 (b) 5
(c) 3 (d) 1
[NDA-2013(1)]

21. What is the value of the minor of the elements 9 in the determinant $\begin{vmatrix} 10 & 19 & 2 \\ 0 & 13 & 1 \\ 9 & 24 & 2 \end{vmatrix}$?
(a) –9 (b) –7
(c) 7 (d) 0
[NDA-2013(1)]

22. The determinant of an orthogonal matrix is:
(a) ±1 (b) 2
(c) 0 (d) ±2
[NDA-2013(1)]

23. The cofactor of the element 4 in the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 8 & 9 \end{vmatrix}$ is
(a) 2 (b) 4
(c) 6 (d) –6
[NDA (II) - 2013]

24. What is the value of the determinant
$\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix}$?
(a) 0 (b) abc
(c) ab + bc + ca (d) abc (a + b + c)
[NDA (II) - 2013]

25. If any two adjacent rows or columns of a determinant are interchanged in position, the value of the determinant
(a) Becomes zero (b) Remains the same
(c) Changes its sign (d) Is doubled
[NDA (I) - 2014]

26. The determinant of an odd order skew-symmetric matrix is always
(a) Zero
(b) One
(c) Negative
(d) Depends on the matrix
[NDA (I) - 2014]

27. One of the roots of $\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0$ is:
(a) abc (b) a+b+c
(c) –(a+b+c) (d) –abc
[NDA-2014(1)]

28. If a ≠ b ≠ c all are positive, then the value of determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is
(a) Non-negative (b) Non-positive
(c) Negative (d) Positive
[NDA (II) - 2014]

29. Consider the following statements
I. Determinant is a square matrix.
II. Determinant is a number associated with asquare matrix.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (II) - 2014]

30. If A and B are two matrices of order 2 and det(A) is 3, det(B) is -1, then find the value of det(3AB)
(a) 3 (b) –9
(c) –27 (d) None of these
[NDA (II) - 2014]

31. If A is invertible matrix, then what is det (A^{-1}) equal to?
(a) det A (b) $\frac{1}{\det A}$
(c) 1 (d) None of these
[NDA (II) - 2014]

32. If $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix} = 0$, then which one of the following is correct?
(a) a/b is one of the cube roots of unity
(b) a/b is one of the cube roots of –1
(c) a is one of the cube roots of unity
(d) b is one of the cube roots of unity
[NDA (II) - 2014]

33. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix}$ =x + iy, where i = √-1, then what is x equal to?
(a) 3 (b) 2
(c) 1 (d) 0
[NDA(II)-2014]

34. Consider the following in respect of two non-singular matrices A and B of same order
I.det (A + B) = det A + det B
II.$(A + B)^{-1} = A^{-1} + B^{-1}$
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2015]

35. The value of $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ is
(a) x + y (b) x – y
(c) xy (d) 1 + x + y
[NDA (I) - 2015]

36. If the value of the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$ is positive, where a $\neq$ b $\neq$ c, then the value of abc.
(a) Cannot be less than 1
(b) Is greater than –8
(c) Is less than –8
(d) Must be greater than 8

[NDA (II) -2015]

37. Consider the following statements in respect of the determinant $\begin{vmatrix} \cos^2\frac{\alpha}{2} & \sin^2\frac{\alpha}{2} \\ \sin^2\frac{\beta}{2} & \cos^2\frac{\beta}{2} \end{vmatrix}$ Where α, β are complementary angles.
I. The value of the determinant is $\frac{1}{\sqrt{2}}\cos\left(\frac{\alpha-\beta}{2}\right)$
II. The maximum value of the determinant is $\frac{1}{\sqrt{2}}$.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (II) - 2015]

38. If A is an invertible matrix of order n and k is any positive real number, then the value of $[\det(kA)]^{-1}$ det A is:
(a) k^{-n} (b) k^{-1}
(c) k^{n} (d) nk

[NDA (II) - 2015]

39. If a, b and c are real numbers, then the value of the determinant $\begin{vmatrix} 1-a & a-b-c & b+c \\ 1-b & b-c-a & c+a \\ 1-c & c-a-b & a+b \end{vmatrix}$ is
(a) 0 (b) (a – b) (b – c) (c – a)
(c) $(a + b + c)^2$ (d) $(a + b + c)^3$

[NDA (II) - 2015]

40. If A is an orthogonal matrix of order 3 and B= $\begin{vmatrix} 1 & 2 & 3 \\ -3 & 0 & 2 \\ 2 & 5 & 0 \end{vmatrix}$, then which of the following is/are correct?
1. |AB|= ±47
2. AB= BA
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2015(2)]

41. Which of the following determinants have value 'zero'?
1. $\begin{vmatrix} 41 & 1 & 5 \\ 79 & 7 & 9 \\ 29 & 5 & 3 \end{vmatrix}$
2. $\begin{bmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{bmatrix}$
3. $\begin{bmatrix} 0 & c & b \\ -c & 0 & a \\ -b & -a & 0 \end{bmatrix}$
Select the correct answer using the codes given below:
(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3

[NDA (I) - 2016]

42. The system of linear equations kx + y + z = 1, x + ky + z = 1 and x + y + kz =1 has a unique solution under which one of the following conditions?
(a) k $\neq$ 1 and k $\neq$ –2 (b) k $\neq$ 1 and k $\neq$2
(c) k $\neq$ –1 and k $\neq$ –2 (d) k $\neq$ –1 and k $\neq$2

[NDA-2016(1)]

43. If A is a square matrix of order 3 and det A = 5, then what is det $[(2A)^{-1}]$ equal to?
(a) $\frac{1}{10}$ (b) $\frac{2}{5}$
(c) $\frac{8}{5}$ (d) $\frac{1}{40}$

[NDA (II) - 2016]

Direction (for next two):
Consider the following for the next two items that follow:
Let $ax^3 + bx^2 + cx + d$
$\begin{vmatrix} x+1 & 2x & 3x \\ 2x+3 & x+1 & x \\ 2-x & 3x+4 & 5x-1 \end{vmatrix}$, then

44. What is the value of c?
(a) –1 (b) 34
(c) 35 (d) 50

[NDA (II) - 2016]

45. What is the value of a + b + c + d?
(a) 62 (b) 63
(c) 65 (d) 68

[NDA (II) - 2016]

46. Which of the following are correct in respect of the system of equations x + y + z = 8, x – y + 2z = 6 and 3x – y + 5z = k?
1.They have no solution, if k = 15
2.They have infinitely many solutions, if k = 20.
3.They have unique solution, if k =25
Select the correct answer using the codes given below:
(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2016]

Direction (for next two):
Consider the following for the next two items that follow:
For the system of linear equations 2x + 3y + 5z = 9, 7x + 3y – 2z = 8 and 2x + 3y + λz = μ.

47. Under what condition does the above system of equations have infinitely many solutions?
(a) λ =5 and μ $\neq$9 (b) λ = 5 and μ = 9
(c) λ = 9 and μ $\neq$ 5 (d) λ = 9 and μ $\neq$ 5

[NDA (II) - 2016]

48. Under what condition does the above system of equation have unique solution?
(a) λ =5 and μ = 9
(b) λ $\neq$ 5 and μ = 7
(c) λ $\neq$ 5 and μ has any real value
(d) λ has any real value of μ $\neq$ 9

[NDA (II) - 2016]

49. What is the value of the determinant?
$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+xyz & 1 \\ 1 & 1 & 1+xyz \end{vmatrix}$?
(a) 1 + x + y + z (b) 2xyz
(c) $x^2 y^2 z^2$ (d) $2x^2 y^2 z^2$

[NDA (I) - 2017]

50. The equations
$x+2y+3z=1$
$2x+y+3z=1$
$5x=5y+9z=4$
(a) Have the unique solution
(b) Have infinitely many solution
(c) Are inconsistent
(d) None of the above

[NDA (I) - 2017]

51. If $\begin{vmatrix} x & y & 0 \\ 0 & x & y \\ y & 0 & x \end{vmatrix}=0$, then which one of the following is correct?
(a) $\frac{x}{y}$ is one of the cube roots of unity
(b) x is one of the cube roots of unity
(c) y is one of the cube roots of unity
(d) $\frac{x}{y}$ is one of the cube roots of -1.

[NDA (I) - 2017]

52. If $A=\begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and det $(A^3)=125$, then α is equal to:
(a) ±1 (b) ±2
(c) ±3 (d) ±5

[NDA (I) - 2017]

53. If $a \neq b \neq c$, then one value of x which satisfies the equation.
$\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix}=0$ is given by:
(a) a (b) b
(c) c (d) 0

[NDA (I) - 2017]

54. If B is a non-singular matrix and A is a square matrix, then the value of det $(B^{-1}AB)$ is equal to:
(a) det (B) (b) det (A)
(c) det (B^{-1}) (d) det (A^{-1})

[NDA (I) - 2017]

55. Consider the set A of all matrices of order 3×3 with entries 0 or 1 only. Let B be the subset of A consisting of all matrices whose determinant is 1. Let C be the subset of A consisting of all matrices whose determinant is –1. Then which one of the following is correct?
(a) C is empty
(b) B has as many elements as C
(c) $A=B\cup C$
(d) B has thrice as many elements as C

[NDA-2017(1)]

56. The value of the determinants $\begin{vmatrix} \cos^2\frac{\theta}{2} & \sin^2\frac{\theta}{2} \\ \sin^2\frac{\theta}{2} & \cos^2\frac{\theta}{2} \end{vmatrix}$ for all values of θ, is:
(a) 1 (b) $\cos\theta$
(c) $\sin\theta$ (d) $\cos2\theta$

[NDA (II) - 2017]

57. The system of equations $kx+y+z=1$, $x+ky+z=k$ and $x+y+kz=k^2$ has no solution if k equals:
(a) 0 (b) 1
(c) –1 (d) –2

[NDA (II) - 2017]

58. If $p+q+r=a+b+c=0$, then the determinant $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$ equal to:
(a) 0 (b) 1
(c) pa + qb + rc (d) pa + qb + rc + a + b + c

[NDA (II) - 2017]

59. The value of the determinant:
$\begin{vmatrix} 1-\alpha & \alpha-\alpha^2 & \alpha^2 \\ 1-\beta & \beta-\beta^2 & \beta^2 \\ 1-\gamma & \gamma-\gamma^2 & \gamma^2 \end{vmatrix}$ is equal to:
(a) $(\alpha-\beta)(\beta-\gamma)(\alpha-\gamma)$
(b) $(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$
(c) $(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)(\alpha+\beta+\gamma)$
(d) 0

[NDA (II) - 2017]

60. If A, B and C are the angles of a triangle and $\begin{vmatrix} 1 & 1 & 1 \\ 1+\sin A & 1+\sin B & 1+\sin C \\ \sin A+\sin^2 A & \sin B+\sin^2 B & \sin C+\sin^2 C \end{vmatrix}=0$, then which one of the following is correct?
(a) The triangle ABC is isosceles
(b) The triangle ABC is equilateral
(c) The triangle ABC is scalene
(d) No conclusion can be drawn with regard to the nature of the triangle

[NDA (II) - 2018]

61. What is the determinant of the matrix $\begin{vmatrix} x & y & y+z \\ z & x & z+x \\ y & z & x+y \end{vmatrix}$?
(a) (x–y) (y–z) (z–x) (b) (x–z) (z–x)
(c) (y–z) (z–x) (d) $(z-x)^2(x+y+z)$

[NDA (II) - 2018]

62. If u, v and w (all positive) are the p^th^, q^th^, and r^th^ terms of a GP, the determinant of the matrix $\begin{vmatrix} \ln u & p & 1 \\ \ln v & q & 1 \\ \ln w & r & 1 \end{vmatrix}$ is:
(a) 0 (b) 1
(c) (p–q) (q–r) (r–p) (d) $\ln u\times\ln v\times\ln w$

[NDA (II) - 2018]

63. If $a+b+c=0$, then one of the solution of $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix}=0$ is
(a) $x=a$ (b) $x=\sqrt{\frac{3(a^2+b^2+c^2)}{2}}$
(c) $x=\sqrt{\frac{2(a^2+b^2+c^2)}{3}}$ (d) $x=0$

[NDA (II) - 2018]

64. Which one of the following factors does the expansions of the determinant $\begin{pmatrix} x & y & 3 \\ x^2 & 5y^3 & 9 \\ x^3 & 10y^5 & 27 \end{pmatrix}$ contain?
(a) x–3 (b) x–y
(c) y–3 (d) d–3y

[NDA (II) - 2018]

65. The system of equation
$2x+y-3z = 5$
$3x-2y+2z = 5$ and
$5x-3y-z = 16$
(a)Is inconsistent
(b)Is consistent, with a unique solution
(c)Is consistent, with infinitely many solution
(d)Has its solution lying along x-axis in three dimensional space.

[NDA (II) - 2018]

66. For a square matrix A, which of the following properties hold?
I.$(A^{-1})^{-1} = A$
II.$\det(A^{-1}) = \dfrac{1}{\det A}$
III.$(\lambda A)^{-1} = \lambda A^{-1}$ where λ is a scalar
Select the correct answer using the code given below:
(a) I and II only (b) II and III only
(c) 1 and III only (d) 1,2 and 3 only

[NDA (II) - 2018]

67. If $\begin{vmatrix} x & -3i & 1 \\ y & 1 & i \\ 0 & 2i & -i \end{vmatrix} = 6+11i$ then what are the value of x and y respectively?
(a) - 3, 4 (b) 3, 4
(c) 3, - 4 (d) - 3, - 4

[NDA (I) - 2019]

68. A is a square matrix of order 3 such that its determinant is 4. What is the determinant of its transpose?
(a) 64 (b) 36
(c) 32 (d) 4

[NDA (I) - 2019]

69. If A is a square matrix of order n > 1, then which one of the following is correct?
(a) $\det(-A) = \det(A)$
(b) $\det(-A) = (-1)^n \det(A)$
(c) $\det(-A) = -\det(A)$
(d) $\det(-A) = n \det(A)$

[NDA (I) - 2019]

Direction for next two questions:
Let A and B be (3×3) matrices with det A = 4 and det B = 3

70. What is det (2AB) equal to?
(a) 96 (b) 72
(c) 48 (d) 36

[NDA (I) - 2019]

71. What is det $(3AB^{-1})$ equal to?
(a) 12 (b) 18
(c) 36 (d) 48

[NDA (I) - 2019]

72. What is the value of determinant
$\begin{vmatrix} 1! & 2! & 3! \\ 2! & 3! & 4! \\ 3! & 4! & 5! \end{vmatrix}$?
(a) 0 (b) 12
(c) 24 (d) 36

[NDA (II) - 2019]

73. What are the values of x that satisfy equation
$$\begin{vmatrix} x & 0 & 2 \\ 2x & 2 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 3x & 0 & 2 \\ x^2 & 2 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 0?$$
(a) $-2\pm\sqrt{3}$ (b) $-1\pm\sqrt{3}$
(c) $-1\pm\sqrt{6}$ (d) $-2\pm\sqrt{6}$

[NDA (II) - 2019]

74. If x + a + b + c = 0, then what is the value of
$\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix}$?
(a) 0 (b) $(a + b + c)^2$
(c) $a^2 + b^2 + c^2$ (d) a + b + c – 2

[NDA (II) - 2019]

75. Let p, q, r be three distinct positive real numbers, If D = $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix}$, then which one of the following is correct ?
(a) $D < 0$ (b) $D \le 0$
(c) $D > 0$ (d) $D \ge 0$

[NDA 2020]

76. What is the value of the determinant
$\begin{vmatrix} i & i^2 & i^3 \\ i^4 & i^6 & i^8 \\ i^9 & i^{12} & i^{15} \end{vmatrix}$ where $i = \sqrt{-1}$?
(a) 0 (b) –2
(c) $4i$ (d) $-4i$

[NDA 2020]

77. If the determinant $\begin{vmatrix} x & 1 & 3 \\ 0 & 0 & 1 \\ 1 & x & 4 \end{vmatrix}$ =0, then what is x =
(a) –2 or 2 (b) –3 or 3
(c) –1 or 1 (d) 3 or 4

[NDA (I) - 2021]

78. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & x(x+1) \\ 3x(x-1) & 2(x-1)(x-2) & x(x+1)(x-1) \end{vmatrix}$ then what is $f(-1)+f(0)+f(1)=$
(a) 0 (b) 1
(c) 100 (d) –100

[NDA (I) - 2021]

79. The element in the i th row and the j th column of a determinant of third order is equal to $2(i + j)$. What is the value of determinant?
(a) 0 (b) 2
(c) 4 (d) 6

[NDA (I) - 2021]

80. With the numbers 2,4,6,8 all the possible determinants with these four different elements are constructed. What is the sum of values of all such determinants?
(a) 128 (b) 64
(c) 32 (d) 0

[NDA (I) - 2021]

81. If Δ is the number of the determinant
$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$
then what is the value of the following determinant?

$$\begin{vmatrix} pa_1 & b_1 & qc_1 \\ pa_2 & b_2 & qc_2 \\ pa_3 & b_3 & qc_3 \end{vmatrix}$$

[$p \neq 0$ or 1 , $q \neq 0$ or 1]

(a) $p\Delta$ (b) $q\Delta$
(c) $(p + q)\Delta$ (d) $pq\Delta$

[NDA (I) - 2021]

82. If a + b + c = 4 and ab + bc + ca = 0, then what is the value of the following determinant?

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

(a) 32 (b) −64
(c) −128 (d) 64

[NDA (I) - 2021]

83. If $a_1, a_2, a_3, ...a_9$ are in GP, then what is the value of the following determinant?

$$\begin{vmatrix} \ln a_1 & \ln a_2 & \ln a_3 \\ \ln a_4 & \ln a_5 & \ln a_6 \\ \ln a_7 & \ln a_8 & \ln a_9 \end{vmatrix}$$

(a) 0 (b) 1
(c) 2 (d) 4

[NDA (I) - 2021]

84. If $A = \begin{vmatrix} p & q \\ r & s \end{vmatrix}$ where p, q, r and s are any four different prime numbers less than 20, what is the maximum value of the determinant?

(a) 215 (b) 311
(c) 317 (d) 323

[NDA (I) - 2021]

85. If A and B are square matrices of order 2 such that det(AB) = det(BA), then which one of the following is correct ?

(a) A must be a unit matrix
(b) B must be a unit matrix
(c) Both A and B must be a unit matrix
(d) A and B need not be a unit matrix

[NDA (I) - 2021]

86. If $\Delta = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ then what is $\begin{vmatrix} 3d+5g & 4a+7g & 6g \\ 3e+5h & 4b+7h & 6h \\ 3f+5i & 4c+7i & 6i \end{vmatrix}$ equal to?

(a) Δ (b) 7Δ
(c) 72Δ (d) -72Δ

[NDA (II) - 2021]

87. If $\begin{vmatrix} a & -b & a-b-c \\ -a & b & -a+b-c \\ -a & -b & -a-b+c \end{vmatrix} - kbac = 0$

($a \neq 0, b \neq 0, c \neq 0$)

then what is the value of k?

(a) –4 (b)–2
(c) 2 (d) 4

[NDA (II) - 2021]

88. If $x = \frac{a}{b-c}, y = \frac{b}{c-a}, z = \frac{c}{a-b}$ then when is the value of the following?

$$\begin{vmatrix} 1 & -x & x \\ 1 & 1 & -y \\ 1 & z & 1 \end{vmatrix}$$

(a) 0 (b) 1
(c) abc (d) ab + bc + ca

[NDA (II) - 2021]

89. For what values of k is the system of equations $2k^2x + 3y - 1 = 0$, $7x - 2y + 3 = 0$, $6kx + y + 1 = 0$ consistent?

(a) $\frac{3 \pm \sqrt{11}}{10}$ (b) $\frac{21 \pm \sqrt{161}}{10}$
(c) $\frac{3 \pm \sqrt{7}}{10}$ (d) $\frac{4 \pm \sqrt{11}}{10}$

[NDA (II) - 2021]

90. What is the value of following determinant

$$\begin{vmatrix} \cos C & \tan A & 0 \\ \sin B & 0 & -\tan A \\ 0 & \sin B & \cos C \end{vmatrix}$$

(a) –1 (b) 0
(c) 2 tanA sinB sinC (d) –2 tanA sinB sinC

[NDA (II) - 2021]

91. If $\Delta_1 = \begin{vmatrix} 1 & p & q \\ 1 & q & r \\ 1 & r & p \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} 1 & 1 & 1 \\ q & r & p \\ r & p & q \end{vmatrix}$ where $p \neq q \neq r$, then $\Delta_1 + \Delta_2$ is:

(a) 0
(b) always positive
(c) always negative
(d)positive if p, q, r are positive else negative.

[NDA (I) - 2022]

92. If (a–b) (b–c) (c–a) = 2 and abc = 6, then what is the value of $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$?

(a) 3 (b) 12
(c) 14 (d) 15

[NDA (I) - 2022]

93. Under which of the following conditions does the determent $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ vanish?

1.$a + b + c = 0$
2.$a^3 + b^3 + c^3 = 3abc$
3.$a^2 + b^2 + c^2 - ab - bc - ca = 0$

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (I) - 2022]

94. Consider the determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

If $a_{13} = yz$, $a_{23} = zx$, $a_{33} = xy$ and the minors of a_{13}, a_{23}, a_{33} are respectively (z–y), (z–x), (y–x) then what is the value of A?

(a) (x–y) (z–x) (y–x)
(b) (x–y) (y–z) (x–z)
(c) (x–y) (z–x) (y–z) (x+y+z))
(d) (xy+yz+zx)(x+y+z)

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:

Let Δ be the determinant of a matrix, A, where A= $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ and C_{11}, C_{12}, C_{13} be the cofactors of a_{11}, a_{12}, a_{13} respectively.

95. What is the value of $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$?
(a) 0 (b) 1
(c) Δ (d) $-\Delta$

[NDA 2022 (II)]

96. What is the value of $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13}$?
(a) 0 (b) 1
(c) Δ (d) $-\Delta$

[NDA 2022 (II)]

97. What is the value of $\begin{vmatrix} a_{21} & a_{31} & a_{11} \\ a_{23} & a_{33} & a_{13} \\ a_{22} & a_{32} & a_{12} \end{vmatrix}$
(a) 0 (b) 1
(c) Δ (d) $-\Delta$

[NDA 2022 (II)]

Consider the following for the next two (02) items that follow:

Let $\Delta(a, b, c, \alpha) = \begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$

98. If $\Delta(a, b, c, \alpha) = 0$ for every $\alpha > 0$, then which one of the following is correct?
(a) a, b, c, are in AP (b) a, b, c are in GP
(c) a, 2b, c are in AP (d) a, 2b, c are in GP

[NDA – 2023 (1)]

99. If $\Delta(7, 4, 2, \alpha) = 0$, then α is a root of which one of the following equations?
(a) $7x^2 + 4x + 2 = 0$ (b) $7x^2 - 4x + 2 = 0$
(c) $7x^2 + 8x + 2 = 0$ (d) $7x^2 - 8x + 2 = 0$

[NDA – 2023 (1)]

100. If a, b, c are in AP, then what is $\begin{vmatrix} x+1 & x+2 & x+3 \\ x+2 & x+3 & x+4 \\ x+a & x+b & x+3 \end{vmatrix}$ equal to?
(a) –1 (b) 0
(c) 1 (d) 2

[NDA – 2023 (1)]

101. The system of linear equations $x + 2y + z = 4$, $2x + 4y + 2z = 8$ and $3x + 6y + 3z = 10$ has:
(a) a unique solution (b) infinite many solution
(c) no solution (d) exactly three solutions

[NDA – 2023 (1)]

102. What is the sum of the roots of the equation $\begin{vmatrix} 0 & x-a & x-b \\ 0 & 0 & x-c \\ x+b & x+c & 1 \end{vmatrix} = 0$?
(a) $a + b + c$ (b) $a - b + c$
(c) $a + b - c$ (d) $a - b - c$

[NDA – 2023 (1)]

103. Let A be a matrix of order 3×3 and $|A| = 4$. If $|2\text{adj}(3A)| = 2^{\alpha}3^{\beta}$, then what is the value of $(\alpha+\beta)$?
(a) 12 (b) 13
(c) 17 (d) 24

[NDA – 2023 (1)]

104. If A, B and C are square matrics of order 3 and $\det(BC) = 2\det(A)$, then what is the value of $\det(2A^{-1}BC)$?
(a)16 (b) 8
(c) 4 (d) 2

[NDA-2023 (2)]

105. Let A be a skew-symmetric matrix of order 3. What is the value of $\det(4A^4) - \det(3A^3) + \det(2A^2) - \det(A) + \det(-I)$ where I is the identity matrix of order 3?
(a) –1 (b) 0
(c) 1 (d) 2

[NDA-2023 (2)]

106. If $\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + e$, then what is the value of e?
(a)–1 (b) 0
(c) 1 (d) 2

[NDA-2023 (2)]

107. If all elements of a third order determinant are equal to 1 or –1, then the value of the determinants is:
(a) 0 only
(b) an even number but not necessarily 0
(c) an odd number
(d) 0, 1 or –1

[NDA-2023 (2)]

108. The value of the determinant of a matrix A of order 3 is 3. If C is the matrix of cofactors of the matrix A, then what is the value of determinant of C^2?
(a) 3 (b) 9
(c) 81 (d) 729

[NDA-2023 (2)]

109. If $A_k = \begin{bmatrix} k-1 & k \\ k-2 & k+1 \end{bmatrix}$, then what is $\det(A_1) + \det(A_2) + \det(A_3) + \ldots + \det(A_{100})$ equal to?
(a) 100 (b) 1000
(c) 9900 (d) 10000

[NDA-2023 (2)]

110. If a, b, c are the sides of triangle ABC, then what is $\begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$ equal to ?
(a) Zero
(b) Area of triangle
(c) Perimeter of triangle
(d) $a^2 + b^2 + c^2$

[NDA – 2024 (1)]

111. For what value of n is the determinant $\begin{vmatrix} C(9,4) & C(9,3) & C(10,n-2) \\ C(11,6) & C(11,5) & C(12,n) \\ C(m,7) & C(m,6) & C(m+1,n+1) \end{vmatrix} = 0$ for every $m > n$?
(a) 4 (b) 5
(c) 6 (d) 7

[NDA – 2024 (1)]

112. If ABC is a triangle, then what is the value of the determinant $\begin{vmatrix} \cos C & \sin B & 0 \\ \tan A & 0 & \sin B \\ 0 & \tan(B+C) & \cos C \end{vmatrix}$?

(a) –1 (b) 0
(c) 1 (d) 3

[NDA – 2024 (1)]

113. Let A and B be matrices of order 3x3. If $|A|=\frac{1}{2\sqrt{2}}$ and $|B|=\frac{1}{729}$, then what is the value of |2B(adj(3A)|?

(a) 27 (b) $\frac{27}{2\sqrt{2}}$
(c) $\frac{27}{2}$ (d) 1

[NDA – 2024 (1)]

114. If in a triangle ABC, $\sin^3A + \sin^3B + \sin^3C = 3\sin A \sin B \sin C$, then what is the value of determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$; where a, b, c are sides of the triangle?

(a) a + b + c (b) ab + bc + ca
(c) (a + b)(b + c)(c + a) (d) 0

[NDA – 2024 (1)]

115. If $\omega = -\frac{1+i\sqrt{3}}{2}$ then what is $\begin{vmatrix} 1+\omega & 1+\omega^2 & \omega+\omega^2 \\ 1 & \omega & \omega^2 \\ \frac{1}{\omega} & \frac{1}{\omega^2} & 1 \end{vmatrix}$ equal to

(a) 0 (b) ω
(c) ω^2 (d) $1 - \omega^2$

[NDA-2024 (2)]

116. If $D_n = \begin{vmatrix} n & 20 & 30 \\ n^2 & 40 & 50 \\ n^3 & 60 & 70 \end{vmatrix}$ then what is the value of $\sum_{n=1}^{4} D_n$?

(a) – 10000 (b) – 10
(c) 10 (d) 10000

[NDA-2024 (2)]

117. If P is a skew-symmetric matrix of order 3, then what is det(P) equal to?

(a) – 1 (b) 0
(c) 1 (d) 3

[NDA-2024 (2)]

118. If $\Delta = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$

and A,B,C,D are the cofactors of the elements a, b, c, d, g respectively, then what is bB + cC – dD – gG equal to?

(a) 0 (b) 1
(c) Δ (d) – Δ

[NDA-2025 (1)]

119. Consider the following statements in respect of the determinant

$$\Delta = \begin{vmatrix} k(k+2) & 2k+1 & 1 \\ 2k+1 & k+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

I. Δ is positive if k > 0.
II. Δ is negative if k < 0.
III. Δ is zero if k = 0.
How many of the statements given above are correct?

(a) None (b) One
(c) Two (d) All three

[NDA-2025 (1)]

120. If $\begin{vmatrix} 2 & 3+i & -1 \\ 3-i & 0 & i-1 \\ -1 & -1-i & 1 \end{vmatrix} = A + iB$

Then what is A + B equal to?

(a) –10 (b) –6
(c) 0 (d) 6

[NDA-2025 (1)]

121. If $A^2 + B^2 + C^2 = 0$, then what is the value of the following?

$$\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix}$$

(a) –1 (b) 0
(c) 1 (d) 2

[NDA-2025 (1)]

122. If ω is a non-real cube root of unity, then what is a root of the following equation?

$$\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$$

(a) x = 0 (b) x = 1
(c) x = ω (d) $x = \omega^2$

[NDA-2025 (1)]

123. What is the value of the determinant of the inverse of the matrix $\begin{bmatrix} -4 & -5 \\ 2 & 2 \end{bmatrix}$?

(a) 1/2 (b) 1
(c) 2 (d) 4

[NDA-2025 (2)]

124. The system of equations 2x – 3y – 5 = 0, 15y – 10x + 50 = 0

(a) has a unique solution
(b) has infinitely many solutions
(c) is inconsistent
(d) is consistent and has exactly two solutions

[NDA-2025 (2)]

125. In obtaining the solution of the system of equations x + y + z = 7, x + 2y + 3z = 16, and x + 3y + 4z = 22 by Cramer's rule, the value of y is obtained by dividing D by D_2, where

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{vmatrix}$$

What is the value of the determinant D_2?

(a) –13 (b) -3
(c) 3 (d) 13

[NDA-2025 (2)]

126. The value of determinant $\begin{vmatrix} a & b & c \\ l & m & n \\ p & q & r \end{vmatrix}$ is equal to

(a) $\begin{vmatrix} a & b & c \\ p & q & r \\ l & m & n \end{vmatrix}$ (b) $\begin{vmatrix} l & m & n \\ a & b & c \\ p & q & r \end{vmatrix}$

(c) $\begin{vmatrix} p & q & r \\ a & b & c \\ l & m & n \end{vmatrix}$ (d) $\begin{vmatrix} a & p & l \\ b & q & m \\ c & r & n \end{vmatrix}$

[NDA-2025 (2)]

127. If *a, b, c* are the sides of a triangle *ABC* and *p* is the perimeter of the triangle, then what is

$$\begin{vmatrix} p+c & a & b \\ c & p+a & b \\ c & a & p+b \end{vmatrix}$$
equal to?
(a) p^3 (b) $2p^3$
(c) $3p^3$ (d) $4p^3$
[NDA-2025 (2)]

Direction for next two :
Let $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

128. What is the value of the determinant of the matrix A^4?
(a) 0 (b) 1
(c) $\cos4\theta - \sin4\theta$ (d) $\cos^2 4\theta - \sin^2 4\theta$
[NDA-2025 (2)]

129. What is $[\text{adj}A]^{-1}$ equal to ?
(a) – A (b) – A^T
(c) A (d) A^T
[NDA-2025 (2)]

130. Consider the following statements:
Statement–I:
If X is an n × n matrix, then
det(mX) = m^n det(X), where m is a scalar.
Statement–II:
If Y is a matrix obtained from X by multiplying any row or column by a scalar m , then det(Y) = m det(X).
Which one of the following is correct in respect of the above statements?
(a) Both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
(b) Both Statement–I and Statement–II are correct but Statement–II does not explain Statement–I.
(c) Statement–I is correct but Statement–II is not correct.
(d) Statement–I is not correct but Statement–II is correct.
[NDA-2025 (2)]

131. Consider the following statements about the matrix
$$M = \begin{bmatrix} 71 & 23 & 48 \\ 57 & 28 & 29 \\ 65 & 17 & 48 \end{bmatrix}$$
Statement–I: The inverse of M does not exist.
Statement–II: M is non-singular.
Which one of the following is correct in respect of the above statements?
(a) Both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
(b) Both Statement–I and Statement–II are correct but Statement–II does not explain Statement–I.
(c) Statement–I is correct but Statement–II is not correct.
(d) Statement–I is not correct but Statement–II is correct.
[NDA-2025 (2)]

ANSWER KEY

1.	d	**2.**	a	**3.**	c	**4.**	a	**5.**	c	**6.**	c	**7.**	a	**8.**	b	**9.**	a	**10.**	a
11.	d	**12.**	a	**13.**	c	**14.**	d	**15.**	a	**16.**	a	**17.**	c	**18.**	b	**19.**	c	**20.**	c
21.	b	**22**	a	**23.**	c	**24.**	a	**25.**	c	**26.**	a	**27.**	c	**28.**	c	**29.**	b	**30.**	c
31.	b	**32.**	b	**33.**	d	**34.**	d	**35.**	c	**36.**	b	**37.**	c	**38.**	a	**39.**	a	**40.**	a
41.	d	**42.**	a	**43.**	d	**44.**	c	**45.**	b	**46.**	a	**47.**	b	**48.**	c	**49.**	c	**50.**	a
51.	d	**52.**	c	**53.**	d	**54.**	b	**55.**	b	**56.**	b	**57.**	d	**58.**	a	**59.**	b	**60.**	a
61.	d	**62.**	a	**63.**	d	**64.**	a	**65.**	b	**66.**	a	**67.**	a	**68.**	d	**69.**	b	**70.**	a
71.	c	**72.**	c	**73.**	d	**74.**	a	**75.**	a	**76.**	d	**77.**	c	**78.**	a	**79.**	a	**80.**	d
81.	d	**82.**	b	**83.**	a	**84.**	c	**85.**	d	**86.**	c	**87.**	a	**88.**	a	**89.**	b	**90.**	b
91.	c	**92.**	b	**93.**	d	**94.**	a	**95.**	c	**96.**	a	**97.**	d	**98.**	b	**99.**	c	**100.**	b
101.	b	**102.**	b	**103.**	b	**104.**	a	**105.**	a	**106.**	b	**107.**	b	**108.**	c	**109.**	d	**110.**	a
111.	c	**112.**	b	**113.**	d	**114.**	d	**115.**	a	**116.**	a	**117.**	b	**118.**	a	**119.**	b	**120.**	b
121.	b	**122.**	a	**123.**	a	**124.**	c	**125.**	b	**126.**	c	**127.**	b	**128.**	b	**129.**	c	**130.**	a
131.	c																		

Solutions

Sol.1. (d)
$$\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}$$
Expanding with respect to R_1,
$= (x + 1) (x^2 + 19x + 70 - x^2 - 18x - 80)$
$-(x + 2) (x^2 + 17x + 42 - x^2 - 15x - 56)$
$+ (x + 4) (x^{2+} + 13x + 30 - x^2 - 35 - 12x)$
$=(x + 1) (x - 10) - (x + 2) (2x - 14) + x$
$(x + 4) (x - 5)$
$= x^2 - 10x + x - 10 - 2x^2 + 14x - 4x + 28$
$+ x^2 - 5x + 4x - 20 = - 2$

Sol.2. (a)
Given determinant is, $\begin{vmatrix} x & 4 & 5 \\ 7 & x & 7 \\ 5 & 8 & x \end{vmatrix} = 0$
Expanding with respect of R_1,
$x(x^2 - 65) - 4 (7x - 35) + 5 (56 - 5x) = 0$
$\Rightarrow x^3 - 56x - 28x + 140 + 280 - 25x = 0$
$\Rightarrow x^3 - 109x + 420 = 0$
If (5, 7) are the roots of the above equation, then
$x^2(x-5) + 5x (x-5) - 84 (x-5)= 0$
$\Rightarrow(x-5) (x^2 + 5x - 84) = 0$
$\Rightarrow(x-5) (x-7) (x+12) = 0 \Rightarrow x=5, 7, -12$

Sol.3. (c)
$kx + 2y = 5$
$3x + y = 1$
$\frac{k}{3} = \frac{2}{1} \Rightarrow k = 6$

Sol.4. (a)

Given, $\begin{vmatrix} x & \alpha & 1 \\ \beta & x & 1 \\ \beta & \gamma & 1 \end{vmatrix}=0$

Use operations $R_1 \to R_2 - R_3 \to R_3 - R_1$

$\begin{vmatrix} x & \alpha & 1 \\ \beta - x & x-\alpha & 0 \\ \beta - x & \gamma-\alpha & 0 \end{vmatrix}=0$

Expand with respect to C_3,

$(\beta - x)(\gamma - \alpha) - (x - \alpha)(\beta - x) = 0$

$\Rightarrow (\beta - x)\{(-\alpha + \gamma) - (x - \alpha)\} = 0$

$\Rightarrow (\beta - x)\{-\alpha + \gamma - x + \alpha\} = 0$

$\Rightarrow (\beta - x)(\gamma - x) = 0$

$\therefore x = \beta, \gamma$

So, root of the given equation is independent of α.

Sol.5. (c)

$\begin{vmatrix} p & -q & 0 \\ 0 & p & q \\ q & 0 & p \end{vmatrix}=0$

Expand with respect or R_1.

$p(p^2 - 0) + q(0 - q^2) + 0 = 0$

$\Rightarrow p^3 - q^3 = 0$

$\Rightarrow (p - q)(p^2 + q^2 + pq) = 0$

$\Rightarrow p-q = 0$, and $p^2 + q^2 + pq = 0$

$p = q$, and $\frac{p^2}{q^2}+1+\frac{pq}{q^2}=0$

$\Rightarrow \left(\frac{p}{q}\right)=1, \text{and} \left(\frac{p}{q}\right)^2+\left(\frac{p}{q}\right)+1=0$

We conclude that $\left(\frac{p}{q}\right)$ is one of the cube roots of unity.

Sol.6. (c)

$\begin{vmatrix} a-b & b+c & a \\ b-c & c+a & b \\ c-a & a+b & c \end{vmatrix} = \begin{vmatrix} a-b & b+c & a+b+c \\ b-c & c+a & a+b+c \\ c-a & a+b & a+b+c \end{vmatrix}$

(Use $C_3 \to C_2 + C_3$)

$=(a+b+c)\begin{vmatrix} a-b & b+c & 1 \\ b-c & c+a & 1 \\ c-a & a+b & 1 \end{vmatrix}$

$=(a+b+c)\begin{vmatrix} a-b & b+c & 1 \\ 2b-a-c & a-b & 0 \\ b+c-2a & a-c & 0 \end{vmatrix}$

(Use $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$)

$=(a+b+c)\{(a-c)(2b-a-c) - (a-b)(b+c-2a)\}$

$=(a+b+c)\{(2ab - a^2 - ac - 2bc + ac + c^2)$

$-ab - ac + 2a^2 + b^2 + bc - 2ab\}$

$=(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$=(a^3 + b^3 + c^3 - abc)$

Sol.7. (a)

equations $3x + 5y = 7$ and $6x + 10y = 18$

here

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

above condition is for no solution.

Sol.8. (b)

$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

Applying $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$

$\begin{vmatrix} a & -b & 0 \\ 0 & b & -c \\ 1 & 1 & 1+c \end{vmatrix} = k$

$a(b+bc+c)+bc=k$

$ab+bc+ca+abc=k$ ………… (1)

but given that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$

or $ab+bc+ca=0$

apply in equation (i)

$k = abc$

Sol.9. (a)

$|AB| = |A||B|$

$|B| = 0$ so $|AB| = 0$

Only Statement 1 is correct

Sol.10. (a)

If two rows of a determinant are identical, then value of the determinant is zero.

Sol.11. (d)

$8(x - 3) + 5(5 - 6) + 1(15 - 6x) = 2$

$x = 8$

Sol.12. (a)

$|B| = 0$ for given matrix B

So $|AB| = |A||B| = 0$

Sol.13. (c)

Let $\Delta = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c2 \end{vmatrix}$

Taking common a, b and c from rows R_1. R_2 and R_3 respectively,

$\Delta = abc\begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$

Again, taking common a, b and c from column C_1, C_2 and C_3 respectively,

$\Delta = a^2b^2c^2\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$

$= a^2 b^2 c^2 (0 + 2 + 2)$

$= 4a^2 b^2 c^2$

Sol.14. (d)

if each element of any one row is multiplied by r than value of determinant is multiplied by r.

Sol.15. (a)

$\begin{vmatrix} x^2 & 1 & y^2+z^2 \\ y^2 & 1 & z^2+x^2 \\ z^2 & 1 & x^2+y^2 \end{vmatrix}$

(Use $C_3 \to C_3 + C_1$)

$\begin{vmatrix} x^2 & 1 & x^2+y^2+z^2 \\ y^2 & 1 & y^2+z^2+x^2 \\ z^2 & 1 & z^2+x^2+y^2 \end{vmatrix}$

after taking $x^2 + y^2 + z^2$ common from 3rd column, column 2 and 3 will be same.

so value of determinant is equal to zero.

Sol.16. (a)

Given that, D is determinant of order 3 and D' is the determinant obtained by replacing the elements of D by their cofactors.

$\therefore$ D' = Cofactor of D

$\Rightarrow$ |D'| = |Cofactor of D|

$\Rightarrow$ |D'| = |adj (D)|

[$\because$ |Cofactor of D| = |adj (D)|]

$\Rightarrow |D'| = |D|^{(3-1)}$ [$\because |adj(A)| = |A|^{n-1}$]

$\Rightarrow |D'| = |D|^2$

$\therefore D' = D^2$

Sol.17. (c)

I. A matrix is only an arrangement of numbers, it has no definite value.

Since, determinant is a value of the matrix. Hence, two determinants of different orders may have the same value.

Sol.18. (b)

Given that, $\begin{vmatrix} 1 & t-1 & 1 \\ t-1 & 1 & 1 \\ 1 & 1 & t-1 \end{vmatrix} = 0$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$\begin{vmatrix} t+1 & t-1 & 1 \\ t+1 & 1 & 1 \\ t+1 & 1 & t-1 \end{vmatrix} = 0$

$\Rightarrow (t+1)\begin{vmatrix} 1 & t-1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & t-1 \end{vmatrix} = 0$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get

$(t+1)\begin{vmatrix} 1 & t-1 & 1 \\ 0 & 2-t & 0 \\ 0 & 2-t & t-2 \end{vmatrix} = 0$

$\Rightarrow (t + 1)(2 - t)(t - 2) = 0$

$\Rightarrow (t + 2)(t - 2)^2 = 0$

$\therefore t = -1, 2$

Sol.19. (c)

Let $\Delta = \begin{vmatrix} m & n & p \\ p & m & n \\ n & p & m \end{vmatrix}$

Applying $C_1 \to C_1 + C_2 + C_3$, we get

$\Delta = \begin{vmatrix} (m+n+p) & n & p \\ (m+n+p) & m & n \\ (m+n+p) & p & m \end{vmatrix}$

$= (m+n+p)\begin{vmatrix} 1 & n & p \\ 1 & m & n \\ 1 & p & m \end{vmatrix}$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get

$\Delta = (m+n+p)\begin{vmatrix} 1 & n & p \\ 0 & m-n & n-p \\ 1 & p-n & m-p \end{vmatrix}$

$= (m + n + p)$

$= (m + n + p)(m^2 + n^2 + p^2 - mn - np - pm)$

$=1/2\,(m + n + p)[(m - n)^2 + (n - p)^2 + (p - m)^2]$

= A linear factor

Sol.20. (c)

for infinite many solutions

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

$\frac{3}{9} = \frac{-1}{-k}$

$k = 3$

Sol.21. (b)

$\begin{vmatrix} 10 & 19 & 2 \\ 0 & 13 & 1 \\ 9 & 24 & 2 \end{vmatrix}$

minor of 9 is $\begin{vmatrix} 19 & 2 \\ 13 & 1 \end{vmatrix} = 19 - 26 = - 7$

Sol.22. (a)

for orthogonal matrix $A^{-1} = A^{T}$

$|A^{-1}| = |A^{T}|$

$\frac{1}{|A|} = |A|$

$|A|^2 = 1$

so $|A| = 1$ or -1

Sol.23. (c)

Cofactor of the element 4

$$= (-1)^{2+1}\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} = -(18-24) = 6$$

Sol.24. (a)

$$\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix}$$

(Use $C_3 \to C_3 + C_1$)

$$\begin{vmatrix} 1 & bc & ab+bc+ca \\ 1 & ca & ab+bc+ca \\ 1 & ab & ab+bc+ca \end{vmatrix}$$

after taking ab + bc + ca common from 3rd column, column 1 and 3 will be same.

so value of determinant is equal to zero.

Sol.25. (c)

If two adjacent rows or adjacent columns interchange their position, then value of determinant will be multiplied by – 1.

Sol.26. (a)

Determinant of odd ordered skew symmetric matrix is always zero.

Sol.27. (c)

$$\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0$$

Apply $C_1 \leftrightarrow C_1 + C_2 + C_3$

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+b & c \\ x+a+b+c & b & x+c \end{vmatrix} = 0$$

$$(x+a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & x+b & c \\ 1 & b & x+c \end{vmatrix} = 0$$

$x + a + b + c = 0$

$x = -(a + b + c)$

Sol.28. (c)

Apply $C_2 \leftrightarrow C_2 + C_3$

Than take ab + bc + ca common from 2nd column.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

[apply $C_1 \to C_1 + C_2 + C_3$ and take common (a + b + c)]

$= (a + b + c)\,[bc - a^2 - b^2 + ab + bc - ca]$

$= (a + b + c)\,[-(a^2 + b^2 + c - ab - bc - ca)]$

$= -\frac{1}{2}(a+b+c)$

$[(a-b)^2 + (b-c)^2 + (c-a)^2]$

= Negative value

Sol.29. (b)

I. determinant is not a square matrix.

II. determinant is a number associated with square matrix.

only statement II is correct.

Sol.30. (c)

A and B are square matrices of order 2.

We know that, $|kA| = k^n|A|$, where n is order of matrix A.

$\therefore |3BA| = 3^2|A||B|$ $\quad (\because |AB| = |A||B|)$

$= 9(-1)(3)$

$= -27$ $\quad (\because |A| = -1, |B| = 3)$

Sol.31. (b)

$\det(A^{-1}) = \frac{1}{\det A}$

Sol.32.

$$\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix} = 0,$$

$a^3 + b^3 = 0$

$a^3 = -b^3$

$\left(\frac{a}{b}\right)^3 = -1$

$\frac{a}{b} = (-1)^{1/3}$

a/b is the one of the cube root of – 1.

Sol.33. (d)

$$\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$$

$6i(3i^2 + 3) + 3i(4i+20) + 1(12-60i) = x + iy$

$0 - 12 + 60i + 12 - 60i = x + iy = 0 + i0$

so $x = 0$

Sol.34. (d)

I. $|A + B| \neq |A| + |B|$

II. $(A+B)^{-1} \neq A^{-1} + B^{-1}$

both statements are wrong.

Sol.35. (c)

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & x & 0 \\ 0 & 0 & y \end{vmatrix}$$

$\begin{bmatrix} \text{apply } R_2 \to R_2 - R_1 \\ \text{and } R_3 \to R_3 - R_1 \end{bmatrix}$

$= 1(xy - 0) = xy$

[expanding along first column]

Sol.36. (b)

Let $\Delta = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$

$= a(bc - 1) - 1(c - 1) + 1(1 - b)$

$= abc - a - b + c + 2$

$\because \Delta > 0$

$\therefore abc - a - b - c + 2 > 0$

$\Rightarrow abc + 2 > a + b + c$ …..(i)

$\because a \neq b \neq c$

$\Rightarrow$ AM of a, b, c > GM of a, b, c

$\Rightarrow \frac{a+b+c}{3} > (abc)^{1/3}$

$\Rightarrow a + b\, c > 3\,(abc)^{1/3}$ ……(ii)

From Eqs. (i) and (ii),

$abc + 2 > 2(abc)^{1/3}$

Now, let $x = (abc)^{1/3}$, then we have

$x^3 + 2 > 3x$

$\Rightarrow x^3 - 3x + 2 > 0$

$\Rightarrow (x-1)^2(x+2) > 0$

$\Rightarrow x + 2 > 0$ $[\because (x-1)^2 > 0]$

$\Rightarrow x > -2$

$\Rightarrow (abc)^{1/3} > -2$

$\Rightarrow abc > -8$

Sol.37. (c)

I. We have, $\Delta = \begin{vmatrix} \cos^2\frac{\alpha}{2} & \sin^2\frac{\alpha}{2} \\ \sin^2\frac{\beta}{2} & \cos^2\frac{\beta}{2} \end{vmatrix}$

$= \cos^2\frac{\alpha}{2}\cos^2\frac{\beta}{2} - \sin^2\frac{\alpha}{2}\sin\frac{\beta}{2}$

$= \left(\cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\right)$

$\left(\cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\right)$

$= \cos\left(\frac{\alpha-\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$

$= \cos\left(\frac{\alpha-\beta}{2}\right)\cos 45°$

$= \frac{1}{\sqrt{2}}\cos\left(\frac{\alpha-\beta}{2}\right)$ $[\because$ $\quad$ $90°]$

II. The maximum value of cos $\frac{\alpha-\beta}{2}$ is 1.

$\therefore$ The maximum value of the determinate is $\frac{1}{\sqrt{2}}$.

Hence, both statements are correct.

Sol.38. (a)

If 'A' is a matrix order 'n' and 'k' is a positive real number then $[\det(kA)]^{-1}\det A = k^{-n}$

Sol.39. (a)

We have, $\Delta = \begin{vmatrix} 1-a & a-b-c & b+c \\ 1-b & b-c-a & c+a \\ 1-c & c-a-b & a+b \end{vmatrix}$

Applying $C_2 \to C_2 + C_3$, we get

$$\Delta = \begin{vmatrix} 1-a & a & b+c \\ 1-b & b & c+a \\ 1-c & c & a+b \end{vmatrix}$$

Now, applying $C_1 \to C_1 + C_2$, $C_3 \to C_3 + C_2$ and taking common a + b + c from C_3, we get

$$\Delta = (a+b+c)\begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix}$$

$\because$ C_1 and C_3 are identical.

Hence, the determinant is 0.

Sol.40. (a)

A is an orthogonal matrix so $|A| = \pm 1$

and |B| of given matrix B is 47

so $|AB| = |A||B| = \pm 47$

but without A matrix we cannot conclude that AB = BA.

so only statement I is correct.

Sol.41. (d)

$$\begin{vmatrix} 41 & 1 & 5 \\ 79 & 7 & 9 \\ 29 & 5 & 3 \end{vmatrix}$$

$(C_3 \to 8C_3 + C_2)$

$$\begin{vmatrix} 41 & 1 & 41 \\ 79 & 7 & 79 \\ 29 & 5 & 29 \end{vmatrix} = 0 \qquad \{\because C_1 = C_3\}$$

(2) $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = \begin{vmatrix} 1 & a+b+c & b+c \\ 1 & a+b+c & c+a \\ 1 & a+b+c & a+b \end{vmatrix}\{c_2\}$

$=(a+b+c)\begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix}=0$

(3) $\begin{vmatrix} 0 & c & b \\ -c & 0 & a \\ -b & -a & 0 \end{vmatrix}=0$ {$\because$ It is a skew symmetric matrix}

Sol.42. (a)
$kx + y + z = 1$
$x + ky + z = 1$
$x + y + kz = 1$
for unique solutions determinant of coefficient matrix should not be zero.

$\begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix}\neq 0$

after adding all rows

$\begin{vmatrix} k+2 & k+2 & k+2 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix}\neq 0$

$(k+2)\begin{vmatrix} 1 & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix}\neq 0$

$k \neq -2$ and $k \neq 1$

Sol.43. (d)
Given, A is a square matrix of order 3 and $|A| = 5$

$\therefore |(2A)^{-1}| = \frac{|A^{-1}|}{8} = \frac{1}{40}$

Sol.44. (c)
Let $\Delta(x) = ax^3 + bx^2 + cx + d$

$\begin{vmatrix} x+1 & 2x & 3x \\ 2x+3 & x+3 & x \\ 2-x & 3x+4 & 5x-1 \end{vmatrix}$

$= \Delta'(x) = 3ax^2 + 2bx + c$

$\begin{vmatrix} 1 & 2x & 3x \\ 2 & x+1 & x \\ -1 & 3x+4 & 5x-1 \end{vmatrix}+\begin{vmatrix} x+1 & 2 & 3x \\ 2x+3 & 1 & x \\ 2-x & 3 & 5x-1 \end{vmatrix}$

$+\begin{vmatrix} x+1 & 2x & 3 \\ 2x+3 & x+1 & 1 \\ 2-x & 3x+4 & 5 \end{vmatrix}$

$\Rightarrow \Delta'(0) = c =$

$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 4 & -1 \end{vmatrix}+\begin{vmatrix} 1 & 2 & 0 \\ 3 & 1 & 0 \\ 2 & 3 & -1 \end{vmatrix}+\begin{vmatrix} 1 & 0 & 3 \\ 3 & 1 & 1 \\ 2 & 4 & 5 \end{vmatrix}$

$\therefore c = (-1) + (-1)(1-6) + 31$
$= -1 + 5 + 31 = 35$

Sol.45. (b)

$\Delta(1) = a + b + c + d = \begin{vmatrix} 2 & 2 & 3 \\ 5 & 2 & 1 \\ 1 & 7 & 4 \end{vmatrix}$

$= 2(8-7) - 2(20-1) + 3(35-2)$
$= 2 - 2(19) + 3(33)$
$= 2 - 38 + 99 = 63$

Sol.46. (a)
Given equations are
$x + y + z = 8$ …(i)
$x - y + 2z = 6$ …(ii)
$3x - y + 5z = k$ …(iii)

Here, $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & -1 & 5 \end{vmatrix} = 1(-5+2)-(5-6)+1(-1+3)$

$= -3 + 1 + 2 = 0$

$D_1 = \begin{vmatrix} 8 & 1 & 1 \\ 6 & -1 & 2 \\ k & -1 & 5 \end{vmatrix} = 8(-5+2)-(30-k)+1(-6+k)$

$= -24 - 30 + 2k - 6 + k = 3k - 60$

$D_2 = \begin{vmatrix} 1 & 8 & 1 \\ 1 & 6 & 2 \\ 3 & k & 5 \end{vmatrix} = 1(30-2k)-8(5-6)+1(k-18)$

$= 30-2k-40+48+k-18 = 20-k$

$D_3 = \begin{vmatrix} 1 & 1 & 8 \\ 1 & -1 & 6 \\ 3 & -1 & k \end{vmatrix} = 1(-k+6)-1(k-18)+8(-1+3)$

$= -k+6-k+18+16 = 40-2k$

For, $k = 15$
$D = 0$ and $D_1 \neq 0$, $D_2 \neq 0$ and $D_3 \neq 0$.
$\therefore$ No solution for $k = 15$
For $k = 20 \Rightarrow D_1 = D_2 = D_3 = 0$ and $D = 0$
$\therefore$ Infinitely many solution for $k = 20$
Since, D = 0'
$\therefore$ System have no unique solution for any value of k.

Sol.47. (b)
We have,
$2x + 3y + 5z = 9$
$7x + 3y - 2z = 8$
And $2x + 3y + \lambda z = \mu$
For infinite solution, $D = 0$
$\therefore 2(3\lambda + 6) - 3(7\lambda + 4) + 5(21 - 6) = 0$
$\Rightarrow 6\lambda + 12 - 21\lambda - 12 + 75 = 0 \Rightarrow \lambda = 5$

And $\begin{vmatrix} 2 & 3 & 9 \\ 7 & 3 & 8 \\ 2 & 3 & \mu \end{vmatrix}=0$

$\Rightarrow 2(3\mu-24)-3(7\mu-16)+9(21-6)=0$
$\Rightarrow 6\mu - 48 - 21\mu + 48 + 135 = 0$
$\therefore \mu = 9$

Sol.48. (c)
For unique solution, $D \neq 0$
$\therefore \lambda \neq 5$ and μ has any real value.

Sol.49. (c)

$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+xyz & 1 \\ 1 & 1 & 1+xyz \end{vmatrix}$

$R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_3$

$\begin{vmatrix} 0 & -xyz & 0 \\ 0 & xyz & -xyz \\ 1 & 1 & 1+xyz \end{vmatrix}$

Expanding along C_1.
$= 2[(-xyz)(-xyz) - xyz \times 0] = x^2 y^2 z^2$.

Sol.50. (a)
$x + 2y + 3z - 1$; $2x+y + 3z = 2$; $5x + 5y + 9z = 4$ have a unique solution, if $|A| \neq 0$.

So, $= \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 5 & 5 & 9 \end{vmatrix} = 1(9-15) - 2[18-15]+[10-3]$

$= 6 - 6 + 15 = 3 \neq 0$.
$\therefore$ The equation have the unique solution.

Sol.51. (d)

$\begin{vmatrix} x & y & 0 \\ 0 & x & y \\ y & 0 & x \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} x+y & y & 0 \\ x+y & x & y \\ x+y & 0 & x \end{vmatrix} = 0$

$C_1 \rightarrow C_1 + C_2 + C_3$

$\Rightarrow (x+y)\begin{vmatrix} 1 & y & 0 \\ 1 & x & y \\ 1 & 0 & x \end{vmatrix} = 0$

$\Rightarrow (x+y)\begin{vmatrix} 0 & y-x & -y \\ 0 & x & y-x \\ 1 & 0 & x \end{vmatrix} = 0$

$R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_3$
$\Rightarrow (x+y)[(y-x)^2 + xy] = 0$
{expanding along C_3}
$\Rightarrow (x + y)(x^2 - xy + y^2) = 0$
$\Rightarrow x^3 + y^3 = 0$
$\Rightarrow x^3 = -y^3 \Rightarrow \left(\frac{x}{y}\right)^3 = -1$

$\therefore \left(\frac{x}{y}\right)$ is one of the cube roots of (-1)

Sol.52. (c)
Who know that, $|A^n| = |A|^n$
$\Rightarrow \det(A^3) = (\det A)^3$

$\Rightarrow 125 = \begin{vmatrix} \alpha & 2 \\ 2 & \alpha \end{vmatrix}^3$

$\Rightarrow (\alpha^2 - 4)^3 = 125$
$\Rightarrow \alpha^2 - 4 = 5$
$\Rightarrow \alpha^2 = 9$
$\alpha = \pm 3$.

Sol.53. (d)
We know that the determinant of a skew – symmetric matrix of odd order is zero.

Clearly, $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix}$ can be a skew symmetric.

Matrix only when $x = 0$

Sol.54. (b)
$\det(B^{-1}AB) = |B^{-1}|.|A|.|B|$
$= |A|\,|B^{-1}|.\,|B|$
$= |A|$

Sol.55. (b)

Let $B = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$ ……….(i)

here $|B| = 1$
we have three rows, by changing adjacent rows we can get $3! = 6$ matrices

by changing R_1 and R_2 $\Rightarrow \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix}$ ….(ii)

by changing R_2 and R_3 $\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}$ ….(iii)

and so on (iv), (v) and (vi).
Out of these 6 matrices determinant of 3 matrices is 1 and other 3 is – 1.
so there will be same elements in B and C.

Sol.56. (b)

$$\begin{vmatrix}\cos^2\frac{\theta}{2} & \sin^2\frac{\theta}{2}\\ \sin^2\frac{\theta}{2} & \cos^2\frac{\theta}{2}\end{vmatrix} = \cos^4\frac{\theta}{2}-\sin^4\frac{\theta}{2}$$

$$=\left(\cos^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}\right)\left(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}\right)$$

$= \cos\theta$.

Sol.57. (d)

For the system of equations to have no solution.

$$\begin{vmatrix}k & 1 & 1\\ 1 & k & 1\\ 1 & 1 & k\end{vmatrix} = 0$$

$\Rightarrow k(k^2-1)-1\ (k-1) + 1\ (1-k) = 0$

$\Rightarrow (k-1)\ [k(k+1)-1-1] = 0$

$\Rightarrow (k-1)\ [k^2 + k - 2] = 0$

$\Rightarrow (k-1)\ (k-1)\ (k+2) = 0$

$\Rightarrow k=1, -2$

$\Rightarrow k = -2$

{$\because$ k=1 does not satisfy the given condition.}

Sol.58. (a)

$$\begin{vmatrix}pa & qb & rc\\ qc & ra & pb\\ rb & pc & qa\end{vmatrix} = pa\ (a^2\ rq - bcp^2) - qb\ (q^2\ ac - prb^2) + rc\ (pqc^2 - abr^2)$$

$= a^3\ pqr - abcp^3 - q^3\ abc + pqrb^3 + c^3\ pqr - abcr^3$

$= pqr\ (a^3 + b^3 + c^3) - abc\ (p^3 + q^3 + r^3)$

Pqr (3abc) – abc (3pqr) {$\because a + b + c = p + q + r = 0$} = 0

Sol.59. (b)

$$\begin{vmatrix}1-\alpha & \alpha-\alpha^2 & \alpha^2\\ 1-\beta & \beta-\beta^2 & \beta^2\\ 1-\gamma & \gamma-\gamma^2 & \gamma^2\end{vmatrix}$$

$$=\begin{vmatrix}1-\alpha & \alpha-\alpha^2 & \alpha^2\\ 1-\beta & \beta-\beta^2 & \beta^2\\ 1-\gamma & \gamma-\gamma^2 & \gamma^2\end{vmatrix} C_2 \rightarrow C_1 + C_2 + C_3$$

$$=\begin{vmatrix}1 & \alpha & \alpha^2\\ 1 & \beta & \beta^2\\ 1 & \gamma & \gamma^2\end{vmatrix} C_2 \rightarrow C_2 \rightarrow C_3$$

$$=\begin{vmatrix}0 & \alpha-\beta & \alpha^2-\beta^2\\ 0 & \beta-\gamma & \beta^2-\gamma^2\\ 1 & \gamma & \gamma^2\end{vmatrix} \begin{matrix}R_1 \rightarrow R_1 \rightarrow R_2\\ R_2 \rightarrow R_2 \rightarrow R_3\end{matrix}$$

$$= (\alpha-\beta)\ (\beta-\gamma)\begin{vmatrix}0 & 1 & \alpha+\beta\\ 0 & 1 & \beta+\gamma\\ 1 & \gamma & \gamma^2\end{vmatrix}$$

Expanding along C_1,

$= (\alpha - \beta)\ (\beta - \gamma)\ [(\beta+\gamma) - (\alpha + \beta)]$

$= (\alpha - \beta)\ (\beta - \gamma)\ (\gamma-\alpha)$

Sol.60. (a)

$R_1 \rightarrow R_1 - R_2$, $R_3 \rightarrow R_3 - R_2$

$$\begin{vmatrix}-\sin A & -\sin B & -\sin C\\ 1+\sin A & 1+\sin B & 1+\sin C\\ \sin^2 A-1 & \sin^2 B-1 & \sin^2 C-1\end{vmatrix}$$

$R_2 \rightarrow R_2 + R_1$

$$\Rightarrow \begin{vmatrix}-\sin A & -\sin B & -\sin C\\ 1 & 1 & 1\\ -\cos^2 A & -\cos^2 B & -\cos^2 C\end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix}\sin A & \sin B & \sin C\\ 1 & 1 & 1\\ \sin^2 A & \sin^2 B & \sin^2 C\end{vmatrix} = 0\ (R_3 \rightarrow R_3 + R_2)$$

$\Rightarrow$If any two angles are equal than two columns will be same than det. is zero.

Sol.61. (d)

$$\begin{vmatrix}x & y & y+z\\ z & x & z+x\\ y & z & x+y\end{vmatrix} = R_1 \rightarrow R_1 + R_2 + R_3$$

$$\begin{vmatrix}x+y+z & x+y+z & 2(x+y+z)\\ z & x & z+x\\ y & z & x+y\end{vmatrix}$$

$$=(x+y+z)\begin{vmatrix}1 & 1 & 2\\ z & x & z+x\\ y & z & x+y\end{vmatrix}$$

$=(x+y+z)\ (z-x)^2$, (replacing z by x)

Sol.62. (a)

$$\begin{vmatrix}\log u & p & 1\\ \log v & q & 1\\ \log w & r & 1\end{vmatrix} = 0$$

($\because$ u, v, w are in G.P. $\Rightarrow$ log u, log v, log w are in A.P.)

Sol.63. (d)

$R_1 \rightarrow R_1 + R_2 + R_3$

$x = a + b + c = 0$

Sol.64. (a)

$C_1 \rightarrow C_1 \rightarrow C_3$

$$\begin{vmatrix}x-3 & y & 3\\ x^2-9 & 5y^3 & 9\\ x^3-27 & 10y^5 & 27\end{vmatrix}$$

$$(x-3)\begin{vmatrix}1 & y & 3\\ x+3 & 5y^3 & 9\\ x^2+9-34 & 10y^5 & 27\end{vmatrix}$$

Sol.65. (b)

$$\text{Since}\begin{vmatrix}2 & 1 & -3\\ 3 & -2 & 2\\ 5 & -3 & -1\end{vmatrix} = 26 \neq 0$$

$\Rightarrow$System is consistent with unique solution.

Sol.66. (a)

Statement I and II are correct.

Statement III is incorrect because

$(\lambda A)^{-1} = \dfrac{1}{\lambda} A^{-1}, \lambda \neq 0$

Sol.67. (a)

$x(- i + 2) - y(- 3 -2i) = 6 + 11i$

$2x + 3y = 6$

$- x + 2y = 11$

by solving these equations y = 4 and x = -3

Sol.68. (d)

because $|A^T| = |A|$

Sol.69. (b)

by property $|kA| = k^n\ |A|$

Sol.70. (a)

$|kAB| = k^n|A||B| = 2^3(4)(3) = 96$

Sol.71. (c)

$|kAB^{-1}| = k^n|A||B^{-1}| = 3^3(4)(1/3) = 36$

Sol.72. (c)

take 2! common from 2nd row and 3! from 3rd row

$$2!3!\begin{vmatrix}1 & 2 & 6\\ 1 & 3 & 12\\ 1 & 4 & 20\end{vmatrix} = 2\times6\times2 = 24$$

Sol.73. (d)

$$\begin{vmatrix}x & 0 & 2\\ 2x & 2 & 1\\ 1 & 1 & 1\end{vmatrix} + \begin{vmatrix}3x & 0 & 2\\ x^2 & 2 & 1\\ 0 & 1 & 1\end{vmatrix} = 0?$$

$$\begin{vmatrix}3x+x & 0 & 2\\ x^2+2x & 2 & 1\\ 1 & 1 & 1\end{vmatrix} = \begin{vmatrix}4x & 0 & 2\\ x^2+2x & 2 & 1\\ 1 & 1 & 1\end{vmatrix} = 0$$

by soving this determinant

$x^2 + 4x - 2 = 0$

$x = -2 \pm \sqrt{6}$

Sol.74. (a)

$R_1 \rightarrow R_1 + R_2 + R_3$

then take x + a + b + c common

and value of this expression is given zero.

Sol.75. (a)

Expand about first row

$p(rq-p^2) - q(q^2-pr) + r(pq-r^2)$

$= 3pqr - p^3 - q^3 - r^3$

$= -(p + q + r)(p^2 + q^2 + r^2 - pq - qr - pr)$

$= -(p+q+r)\dfrac{1}{2}\left[(p-q)^2+(q-r)^2+(r-p)^2\right]$

given that p, q, r be three distinct positive real numbers

so this expression is always negative.

Sol.76. (d)

$$\begin{vmatrix}i & i^2 & i^3\\ i^4 & i^6 & i^8\\ i^9 & i^{12} & i^{15}\end{vmatrix} = \begin{vmatrix}i & -1 & -i\\ 1 & -1 & 1\\ i & 1 & -i\end{vmatrix}$$

$R_1 \rightarrow R_1 - R_3$

$$\begin{vmatrix}0 & -2 & 0\\ 1 & -1 & 1\\ i & 1 & -i\end{vmatrix} = -4i$$

Sol.77. (c)

$$\begin{vmatrix}x & 1 & 3\\ 0 & 0 & 1\\ 1 & x & 4\end{vmatrix} = 0$$

expand with second row

$0 + 0 - 1(x^2 - 1) = 0$

x = 1 or – 1

Sol.78. (a)

$$f(x) = \begin{vmatrix}1 & x & x+1\\ 2x & x(x-1) & x(x+1)\\ 3x(x-1) & 2(x-1)(x-2) & x(x+1)(x-1)\end{vmatrix}$$

f(–1) = 0 [because 3rd column zero]

f(0) = 0 [because 2nd row zero]

f(1) = 0 [because 3rd row zero]

so f(–1) + f(0) + f(1) = 0

Sol.79. (a)

The element in the i th row and the j th column of a determinant of third order is equal to $2(\ i + j\ )$

$$\begin{vmatrix}4 & 6 & 8\\ 6 & 8 & 10\\ 8 & 10 & 12\end{vmatrix} = -16 + 48 - 32 = 0$$

Sol.80. (d)

take any determinant from 2,4,6,8

$$\begin{vmatrix} 2 & 4 \\ 6 & 8 \end{vmatrix} = 8$$

interchange both row

$$\begin{vmatrix} 6 & 8 \\ 2 & 4 \end{vmatrix} = -8$$

sum of both determinant is zero.
here is 4 elements so total 24 determinant can be made, and sum in pair will be zero.

Sol.81. (b)

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \Delta$$

$$\begin{vmatrix} pa_1 & b_1 & qc_1 \\ pa_2 & b_2 & qc_2 \\ pa_3 & b_3 & qc_3 \end{vmatrix} = pq\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = pq\Delta$$

Sol.82. (b)

a + b + c = 4, ab + bc + ca = 0
put a = 0, b = 0, c = 4

$$\begin{vmatrix} 0 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 0 \end{vmatrix} = -64$$

Sol.83. (a)

$$\begin{vmatrix} \ln a_1 & \ln a_2 & \ln a_3 \\ \ln a_4 & \ln a_5 & \ln a_6 \\ \ln a_7 & \ln a_8 & \ln a_9 \end{vmatrix}$$

$$= \begin{vmatrix} \ln a & \ln ar & \ln ar^2 \\ \ln ar^3 & \ln ar^4 & \ln ar^5 \\ \ln ar^6 & \ln ar^7 & \ln ar^8 \end{vmatrix}$$

$$= \begin{vmatrix} \ln a & \ln a + \ln r & \ln a + 2\ln r \\ \ln a + 3\ln r & \ln a + 4\ln r & \ln a + 5\ln r \\ \ln a + 6\ln r & \ln a + 7\ln r & \ln a + 8\ln r \end{vmatrix}$$

apply $R_3 = R_3 - R_2$ and $R_2 = R_2 - R_1$
second and third row will be same so determinant is zero.

Sol.84. (c)

prime numbers less than 20 are 2,3,5,7,11,13,17,19
max. value will occur when diagonals are max. and others are min.

$$\begin{vmatrix} 19 & 3 \\ 2 & 17 \end{vmatrix} = 323 - 6 = 317$$

Sol.85. (d)

|AB| = |A||B|=|B||A| = |BA| is correct for all matrices.

Sol.86. (c)

$$\begin{vmatrix} 3d & 4a & 6g \\ 3e & 4b & 6h \\ 3f & 4c & 6i \end{vmatrix} + \begin{vmatrix} 5g & 7g & 0 \\ 5h & 7h & 0 \\ 5i & 7i & 0 \end{vmatrix}$$

det is zero as coloumn third contain all terms as 0.

$$-3 \times 4 \times 6 \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} + 0$$

As you know $|A|^T = |A|$ Interchange rows into coloumns

$$-3 \times 4 \times 6 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + 0$$

$= -72\ \Delta$

Sol.87. (a)

$$\begin{vmatrix} a & -b & a-b-c \\ -a & b & -a+b-c \\ -1 & -1 & -a-b+c \end{vmatrix} - k\,abc = 0$$

$$\begin{vmatrix} 1 & -1 & 1-1-1 \\ -1 & 1 & -1+1-1 \\ -1 & -1 & -1-1+1 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & -1 \end{vmatrix}$$

–k (1)=0

$$\Rightarrow k = \begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

(1)[–1–1] – (–1) [–1–1] + (–1) [1+1]
(–2)+(0)+(–2) = k
⇒ k = – 4

Sol.88. (a)

$$x = \frac{a}{b-c},\ y = \frac{b}{c-a},\ \tau = \frac{c}{a-b}$$

$$\text{T.F.} \begin{vmatrix} 1 & -x & x \\ 1 & 1 & -y \\ 1 & \tau & 1 \end{vmatrix}$$

Fix a = 1, b = 2, c = 3

$$\therefore\ x = \frac{1}{-1} ie -1\ \ y = \frac{2}{2} = 1,\ \ z = \frac{3}{-1} ie -3$$

$$\text{So, Det formed is} \begin{vmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 1 & -3 & 1 \end{vmatrix}$$

(1)[1–3]–(1)[1+1] + (–1) [–3 –1]
–2 – 2 + 4
⇒ 4 + 4 = 0

Sol.89. (b)

System of equation is consistent

$$\begin{vmatrix} 2k^2 & 3 & -1 \\ 7 & -2 & 3 \\ 6k & 1 & 1 \end{vmatrix} = 0$$

$2k^2(-5)-3(7-18k)-1(7+12k)=0$
$-10k^2-21+54k-7-12k=0$
$10k^2-42k+28=0$
$5k^2-21k+14=0$

$$K = \frac{21 \pm \sqrt{161}}{10}$$

Sol.90. (b)

cosC tanA sinB – tanA sinB cosC = 0

Sol.91. (c)

$$\Delta_1 \begin{vmatrix} 1 & p & q \\ 1 & q & r \\ 1 & r & p \end{vmatrix} = qp + qr + pr - p^2 - q^2 - r^2$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & 1 \\ q & r & p \\ r & p & q \end{vmatrix} = qr + pr + qp - p^2 - q^2 - r^2$$

$\Delta_1 + \Delta_2 = [2p^2 + 2q^2 + 2r^2 = -2pr - 2qr - 2pr]$
$= [(p-q)^2 + (q-r)^2 + (p-r)^2]$
= always negative

Sol.92. (b)

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = abc\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

=abc (a–b) (b–c) (c-a)
= 2 × 6 =12

Sol.93. (d)

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$a(bc-a^2) - b(b^2-ac) + c(ab-c^2) = 0$
$3abc-a^3-b^3-c^3 = 0$
$3abc - a^3 - b^3 - c^3 = 0$
$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - a) = 0$
any factor may be zer

Sol.94. (a)

if we multiply any row to their cofactors than we will get its determinant
$\Delta = a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33}$
$\Delta = yz(z-y) - zx(z-x) + xy(y-x)$
$\Delta = x^2z - x^2y - z^2x + y^2x + yz(z-y)$
$\Delta = x^2(z-y) - x(z^2 - y^2) + yz(z-y)$
$\Delta = (z-y)[x^2 - x(z+y) + yz]$
$\Delta = (z-y)[yz - xy + x^2 - xz]$
$\Delta = (z-y)[y(z-x) + x(x-z)]$
$\Delta = (z-y)(z-x)(y-x)$
$\Delta = (x-y)(y-z)(z-x)$

Sol.95. (c)

if we multiply any row to their cofactors than we will get its determinant
$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = \Delta$

Sol.96. (a)

if we multiply any row to cofactors of any other row than we will get zero answer.
$a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = 0$

Sol.97. (d)

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$\det(A^T) = \det(A)$

$$\Delta = \begin{vmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{32} & a_{33} \end{vmatrix}$$

Interchange column 1 and 2

$$-\Delta = \begin{vmatrix} a_{21} & a_{11} & a_{31} \\ a_{22} & a_{12} & a_{32} \\ a_{23} & a_{13} & a_{33} \end{vmatrix}$$

Interchange column 2 and 3

$$\Delta = \begin{vmatrix} a_{21} & a_{31} & a_{11} \\ a_{22} & a_{32} & a_{12} \\ a_{23} & a_{33} & a_{13} \end{vmatrix}$$

Interchange row 2 and 3

$$-\Delta = \begin{vmatrix} a_{21} & a_{31} & a_{11} \\ a_{23} & a_{33} & a_{13} \\ a_{22} & a_{32} & a_{12} \end{vmatrix}$$

Sol.98. (b)

$$\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}=0$$

$R_2 \rightarrow R_2 + \alpha R_1 - R_3$

$$\begin{vmatrix} a & b & a\alpha+b \\ 0 & 0 & (b\alpha+c)+\alpha(a\alpha+b) \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}=0$$

$$-[(b\alpha+c)+\alpha(a\alpha+b)][a(b\alpha+c)-b(a\alpha+b)]=0$$

$$[(b\alpha+c)+\alpha(a\alpha+b)][ac-b^2]=0$$

$$[a\alpha^2+2b\alpha+c]\times[ac-b^2]=0$$

$b^2 = ac$

so a, b, c are in GP

Sol.99. (c)

by above solution

$$[a\alpha^2+2b\alpha+c]\times[ac-b^2]=0$$

$$[a\alpha^2+2b\alpha+c]=0$$

Put a = 7, b = 4, c = 2

$$\Delta(a,b,c,\alpha)=[7\alpha^2+8\alpha+2]=0$$

Sol.100. (b)

$$\begin{vmatrix} x+1 & x+2 & x+3 \\ x+2 & x+3 & x+4 \\ x+a & x+b & x+c \end{vmatrix}$$

$R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_2$

$$\begin{vmatrix} x+1 & x+2 & x+3 \\ 1 & 1 & 1 \\ a-1 & b-2 & c-3 \end{vmatrix}$$

$C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_2$

$$\begin{vmatrix} x+1 & 1 & 1 \\ 1 & 0 & 0 \\ a-1 & b-a-1 & c-b-1 \end{vmatrix}$$

= c – b – 1 – (b – a – 1)

= c + a – 2b = 0

Because given that a, b, c are in AP

Sol.101. (b)

x + 2y + z = 4,

2x + 4y + 2z = 8

3x + 6y + 3z = 10

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 6 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

AX = B

|A| = 0

$$\text{adj}(A)=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{adj}(A)B=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

adj(A)B = O so equations have infinite many solutions

Sol.102. (b)

$$\begin{vmatrix} 0 & x-a & x-b \\ 0 & 0 & x-c \\ x+b & x+c & 1 \end{vmatrix}$$

$$=(x+b)(x-a)(x-c)$$

$x = a, -b, c$

Sum of roots = a – b + c

Sol.103. (b)

|2adj (3A)|

$=2^3$|adj (3A)|

$=8|3A|^2$

$=8[3^3|A|]^2$

$=8[27]^2|A|^2$

$=8(3^6)(16) = 2^7 3^6 = 2^\alpha 3^\beta$

so α + β = 13

Sol.104. (a)

$|2A^{-1}BC|$

$= 2^3|A^{-1}BC|$

$=8|A^{-1}||BC|$

$=8|A^{-1}|.2|A| = 16$

Sol.105. (a)

If A is skew symmetric matrix of order than 3 than its |A| = 0

If |A| = 0 than $|A|^n = |A^n| = 0$

$|4A^4| – |3A^3| + |2A^2| - |A| + |–I|$

$= 4^3|A^4| – 3^3|A^3| + 2^3|A^2| – |A| – |I|$

= 0 – 0 + 0 – 0 – 1 = – 1

Sol.106. (b)

$$\begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}=ax^4+bx^3+cx^2+dx+e$$

Put x= 0

$$\begin{vmatrix} 0 & -1 & 3 \\ 1 & 0 & -4 \\ -3 & 4 & 0 \end{vmatrix}=e=0$$

Determinant of skew symmetric matrix is zero.

Sol.107. (b)

Let determinant is $\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$

$$R_2 \leftrightarrow R_2-\frac{p}{a}R_1, R_3 \leftrightarrow R_3-\frac{x}{p}R_2$$

All numbers are 1 or -1 so after applying operation we will get only even numbers

$$\begin{vmatrix} a & b & c \\ 0 & 2\alpha & 2\beta \\ 0 & 2\gamma & 2\delta \end{vmatrix}= \text{an even number}$$

And if we take all elements are same than we will get 0 answer.

Sol.108. (c)

$|adjA| = |A|^2$

So $|adjA| = |C| = 3^2 = 9$

So $|C^2|=|C|^2 = 81$

Sol.109. (d)

$$|A_k|=\begin{vmatrix} k-1 & k \\ k-2 & k+1 \end{vmatrix}=k^2-1-k^2+2k=2k-1$$

$|A_1| + |A_2| + |A_3| + |A_4| + ………….|A_{100}|$

=1 + 3 + 5 +7 + 9 + ……+ 199 = 10000

Sol.110. (a)

$$\begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$$

$=a^2$ (1- $\cos^2$A) –bsinA(bsinA – csinAcosA) + csinA(bsinAcosA – csinA)

$= a^2 \sin^2A – b^2\sin^2A + bc\sin^2A\cos A + bc\sin^2A\cos A – c^2\sin^2A$

$=\sin^2A(a^2 – b^2 – c^2 +2bc\cos A)$

And we know that

$$\cos A=\frac{b^2+c^2-a^2}{2bc} \Rightarrow 2bc\cos A=b^2+c^2-a^2$$

So answer is zero.

Sol.111. (c)

$$\begin{vmatrix} C(9,4) & C(9,3) & C(10,n-2) \\ C(11,6) & C(11,5) & C(12,n) \\ C(m,7) & C(m,6) & C(m+1,n+1) \end{vmatrix}=0$$

We know that

$${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$$

$C_1 \rightarrow C_1 + C_2$

$$\begin{vmatrix} C(9,4) & C(9,3) & C(10,n-2) \\ C(11,6) & C(11,5) & C(12,n) \\ C(m,7) & C(m,6) & C(m+1,n+1) \end{vmatrix}=0$$

$$\begin{vmatrix} C(9,4)+C(9,3) & C(9,3) & C(10,n-2) \\ C(11,6)+C(11,5) & C(11,5) & C(12,n) \\ C(m,7)+C(m,6) & C(m,6) & C(m+1,n+1) \end{vmatrix}=0$$

$$\begin{vmatrix} C(10,4) & C(9,3) & C(10,n-2) \\ C(12,6) & C(11,5) & C(12,n) \\ C(m+1,7) & C(m,6) & C(m+1,n+1) \end{vmatrix}=0$$

Compare column 1 and column 3

n = 6

Sol.112. (b)

$$\begin{vmatrix} \cos C & \sin B & 0 \\ \tan A & 0 & \sin B \\ 0 & \tan(B+C) & \cos C \end{vmatrix}$$

A + B + C = 180

B + C = 180 – C

Tan(B +C) = tan (180- A)

Tan(B + C) = – tanA

$$\begin{vmatrix} \cos C & \sin B & 0 \\ \tan A & 0 & \sin B \\ 0 & -\tan A & \cos C \end{vmatrix}$$

=cosC(tanAsinB) – sinB(tanAcosC) = 0

Sol.113. (d)

|2Badj (3A)|

$=2^3$|B||adj (3A)|

$=8|B||3A|^2$

$=8|B|\ [3^3|A|]^2$

$=8|B|\ [27]^2|\ A|^2$

$$=8\times\frac{1}{729}\times 729\times\frac{1}{8}=1$$

Sol.114. (d)

Formula of algebra

$$x^3+y^3+z^3-3xyz=(x+y+z)\frac{1}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]$$

if $x^3+y^3+z^3=3xyz$

than x + y + z = 0 or

$$\frac{1}{2}\left[(x-y)^2+(y-z)^2+(z-x)^2\right]=0$$

sinA + sinB + sinC can not be zero

so

$$\left[(\sin A-\sin B)^2+(\sin B-\sin C)^2+(\sin C-\sin A)^2\right]=0$$

It is only possible when sinA = sinB = sinC

i.e. A = B = C than a = b = c

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=\begin{vmatrix} a & a & a \\ a & a & a \\ a & a & a \end{vmatrix}=0$$

Sol.115. (a)

$$\begin{vmatrix} 1+\omega & 1+\omega^2 & \omega+\omega^2 \\ 1 & \omega & \omega^2 \\ \frac{1}{\omega} & \frac{1}{\omega^2} & 1 \end{vmatrix}$$

we know that $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$ (from complex number chapter)
so

$$\Rightarrow \begin{vmatrix} 1+\omega & 1+\omega^2 & \omega+\omega^2 \\ 1 & \omega & \omega^2 \\ \frac{\omega^3}{\omega} & \frac{\omega^3}{\omega^2} & 1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} -\omega^2 & -\omega & -1 \\ 1 & \omega & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix}$$

$R_1 \to R_1 + R_3$

$$\Rightarrow \begin{vmatrix} 0 & 0 & 0 \\ 1 & \omega & \omega^2 \\ \omega^2 & \omega & 1 \end{vmatrix} = 0$$

Sol.116. (a)

$$D_n = \begin{vmatrix} n & 20 & 30 \\ n^2 & 40 & 50 \\ n^3 & 60 & 70 \end{vmatrix}$$

$$\sum_{n=1}^{4} D_n = D_1 + D_2 + D_3 + D_4$$

$$= \begin{vmatrix} 1 & 20 & 30 \\ 1^2 & 40 & 50 \\ 1^3 & 60 & 70 \end{vmatrix} + \begin{vmatrix} 2 & 20 & 30 \\ 2^2 & 40 & 50 \\ 2^3 & 60 & 70 \end{vmatrix}$$

$$+ \begin{vmatrix} 3 & 20 & 30 \\ 3^2 & 40 & 50 \\ 3^3 & 60 & 70 \end{vmatrix} + \begin{vmatrix} 4 & 20 & 30 \\ 4^2 & 40 & 50 \\ 4^3 & 60 & 70 \end{vmatrix}$$

$$= \begin{vmatrix} 1+2+3+4 & 20 & 30 \\ 1^2+2^2+3^2+4^2 & 40 & 50 \\ 1^3+2^3+3^3+4^3 & 60 & 70 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & 20 & 30 \\ 30 & 40 & 50 \\ 100 & 60 & 70 \end{vmatrix} = 1000 \begin{vmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 10 & 6 & 7 \end{vmatrix} = -10000$$

Sol.117. (b)
Determinant of odd ordered skew symmetric matrix is always zero.

Sol.118. (a)
We know that if we multiply any row or column with its cofactors, then we get the value of determinant.
From first row
$aA + bB + cC = \Delta$ ……(i)
from first column
$aA + dD + gG = \Delta$ …….(ii)
subtract (i) from (ii)
$bB + cC - dD - gG = 0$

Sol.119. (b)

$$\Delta = \begin{vmatrix} k(k+2) & 2k+1 & 1 \\ 2k+1 & k+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

$R_1 \to R_1 - R_2$

$$\Delta = \begin{vmatrix} k^2-1 & k-1 & 0 \\ 2k+1 & k+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (k-1) \begin{vmatrix} k+1 & 1 & 0 \\ 2k+1 & k+2 & 1 \\ 3 & 3 & 1 \end{vmatrix}$$

Now by solving determinant

$$= (k-1)(k-1)^2 = (k-1)^3$$

Statement 1: for $k > 0$, Δ is not always positive, so statement is wrong.
Statement 2: for $k < 0$, Δ is negative, this statement is correct.
for $k = 0$, Δ is negative, so statement is wrong.
Hence only one statement is correct.

Sol.120. (b)

$$\begin{vmatrix} 2 & 3+i & -1 \\ 3-i & 0 & i-1 \\ -1 & -1-i & 1 \end{vmatrix} = A + iB$$

$$= -(3-i)\left[(3+i)+(-1-i)\right] - (i-1)\left[(-2-2i)+(3+i)\right]$$

$$= -(3-i)[2] - (i-1)[1-i]$$

$= -6 = A + iB$
So $A + B = -6$

Sol.121. (b)
$A^2 + B^2 + C^2 = 0$
i.e. $A = B = C = 0$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Sol.122. (a)
$C_1 \to C_1 + C_2 + C_3$

$$\begin{vmatrix} x+1+\omega+\omega^2 & \omega & \omega^2 \\ x+1+\omega+\omega^2 & x+\omega^2 & 1 \\ x+1+\omega+\omega^2 & 1 & x+\omega \end{vmatrix} = 0$$

$$= \begin{vmatrix} x & \omega & \omega^2 \\ x & x+\omega^2 & 1 \\ x & 1 & x+\omega \end{vmatrix} = x \begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & x+\omega^2 & 1 \\ 1 & 1 & x+\omega \end{vmatrix} = 0$$

$x = 0$

Sol.123. (a)
we know that $\left|A^{-1}\right| = \dfrac{1}{|A|}$

$A = \begin{bmatrix} -4 & -5 \\ 2 & 2 \end{bmatrix}$, $|A| = -8 + 10 = 2$

$|A^{-1}| = ½$

Sol.124. (c)
$2x - 3y - 5 = 0$, $15y - 10x + 50 = 0$
Here $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$, so system of equation has no solution or inconsistent equations.

Sol.125. (b)
According to cramer's rule D_2 is a determinant whose column of coefficient of y replaced by constant column marix.

$$D = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 16 & 3 \\ 1 & 22 & 4 \end{vmatrix}$$

$= 64 - 66 - 28 + 22 + 21 - 16 = -3$

Sol.126. (c)

$$\begin{vmatrix} a & b & c \\ l & m & n \\ p & q & r \end{vmatrix}$$

First replace $R_1 \leftrightarrow R_2$, than replace $R_2 \leftrightarrow R_3$

$$\Rightarrow \begin{vmatrix} p & q & r \\ a & b & c \\ l & m & n \end{vmatrix}$$

Sol.127. (b)

$$\begin{vmatrix} p+c & a & b \\ c & p+a & b \\ c & a & p+b \end{vmatrix}$$

$R_1 \leftrightarrow R_1 - R_2$

$$\begin{vmatrix} p & -p & 0 \\ c & p+a & b \\ c & a & p+b \end{vmatrix} = p \begin{vmatrix} 1 & -1 & 0 \\ c & p+a & b \\ c & a & p+b \end{vmatrix}$$

Now we can expand it
$= p[(p + a)(p + b) - ab + pc + bc - bc]$
$= p[p^2 + ap + bp + cp]$
$= p^2[p + a + b + c] = 2p^3$

Sol.128. (b)

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$|A| = 1$
We know that $|A^4| = |A|^4 = 1$

Sol.129. (c)

$$(adjA)^{-1} = \frac{adj(adjA)}{|adjA|}$$

For 2nd order matrix adj(adjA) = A and |adjA| = |A|
So $(adjA)^{-1} = A$

Sol.130. (a)
Both statements are correct and statement 2 is the correct explanation of statement 1.

Sol.131. (c)

$$M = \begin{bmatrix} 71 & 23 & 48 \\ 57 & 28 & 29 \\ 65 & 17 & 48 \end{bmatrix}$$

If we apply only one elementary operation $C_1 \leftrightarrow C_1 - C_2 - C_3$ then we get 1st column completely zero. so determinant of M is zero and matrix is singular, and inverse of singular matrix does not exist.